

Patient-Aware Machine-Learning Classification of Benign and Malignant Breast Lesions Using Single- and Multi-Source MRI Radiomic Features

Abstract

1 Introduction

Machine learning provides a computational approach for identifying patterns in numerical medical-imaging data and using these patterns to classify new observations. In radiomics-based studies, each lesion is represented by a large number of quantitative features. These features can then be used as inputs to a classification model for distinguishing between predefined outcome classes, such as benign and malignant lesions.

Developing reliable machine-learning models from radiomic data is challenging because the number of candidate features is often large compared with the number of available patients. In this setting, a model may fit patterns that are specific to the development data rather than patterns that generalise to unseen patients. Redundant features, differences in numerical feature scales, missing values, class imbalance, and model complexity can further affect the estimated performance. Therefore, feature selection, careful preprocessing, appropriate validation, and controlled hyperparameter optimisation are important parts of the modelling process.

Radiomic features may be available from multiple imaging sources or acquisition sequences. Each source can be analysed separately, or the corresponding feature columns can be combined to create a larger multi-source representation. Combining sources may provide the classifier with additional information. However, it also increases the dimensionality of the input data without in-

creasing the number of independent patients. Consequently, using more feature sources does not necessarily lead to better classification performance. A systematic comparison is required to determine whether individual sources, selected combinations, or complete feature concatenation provide the most effective input representation.

An additional methodological concern arises when one patient contributes more than one region of interest. If regions from the same patient are distributed between the training and test sets, the model may be evaluated using data that are not fully independent from its training data. This can produce an overly optimistic estimate of performance. Patient-aware data splitting is therefore required so that all regions belonging to the same patient remain in the same cross-validation partition. Model optimisation must also be separated from final evaluation to ensure that the test data do not influence preprocessing, feature selection, or parameter tuning.

In this study, a patient-aware machine-learning framework was developed for binary classification using five sets of pre-extracted MRI radiomic features: apparent diffusion coefficient (ADC), pre-contrast imaging (Pre), two post-contrast phases (Post1 and Post2), and T2-weighted imaging (T2). Each source was evaluated individually and in combination. All 31 non-empty source combinations were examined, including five single-source, ten pairwise, ten triple-source, five quadruple-source, and one complete five-source configuration.

The same computational workflow was applied to every dataset configuration. Three sequential

feature-selection strategies were compared, and the number of retained features was selected automatically within the model-development data. Each feature-selection strategy was evaluated with and without the Synthetic Minority Over-sampling Technique (SMOTE) and combined with ten machine-learning classifiers. Model parameters were optimised using patient-aware nested cross-validation, while all data-dependent preprocessing operations were restricted to the corresponding training partitions.

The primary objective was to determine which feature-source combination, feature-selection strategy, class-balancing condition, and classifier achieved the highest performance for distinguishing benign and malignant breast lesions. The study also examined whether classification performance changed as additional feature sources were combined, whether SMOTE improved model performance, and how consistently individual features were selected across different patient groups.

2 Materials and Methods

2.1 Computational Study Design

A supervised machine-learning framework was developed to distinguish benign from malignant breast lesions using pre-extracted numerical radiomic features. The computational analysis started from tabular feature datasets and did not include image segmentation, image preprocessing, or radiomic feature extraction.

Five radiomic feature sources were evaluated: apparent diffusion coefficient (ADC), pre-contrast imaging (Pre), two post-contrast phases (Post1 and Post2), and T2-weighted imaging (T2). Each source was first evaluated separately. Multi-source datasets were then created by combining the feature columns of corresponding regions of interest (ROIs).

Each row of a dataset represented one ROI. The associated class label indicated whether the ROI was benign or malignant. Some patients contributed more than one ROI. Therefore, the ROI

was considered the prediction unit, whereas the patient was used as the grouping unit during model development and evaluation.

All non-empty combinations of the five feature sources were examined. The total number of dataset configurations was calculated as

$$\binom{5}{1} + \binom{5}{2} + \binom{5}{3} + \binom{5}{4} + \binom{5}{5} = 5 + 10 + 10 + 5 + 1 = 31. \quad (1)$$

The computational experiments were organised into five groups:

1. **Experiment 1: Single-source analysis**, including the five ADC, Pre, Post1, Post2, and T2 datasets;
2. **Experiment 2: Pairwise analysis**, including all ten possible combinations of two feature sources;
3. **Experiment 3: Triple-source analysis**, including all ten possible combinations of three feature sources;
4. **Experiment 4: Quadruple-source analysis**, including all five possible combinations of four feature sources;
5. **Experiment 5: Complete multi-source analysis**, including the combination of all five feature sources.

The same machine-learning workflow was applied to each of the 31 dataset configurations. The workflow included dataset validation, training-fold preprocessing, feature selection, automatic selection of the feature-set size, optional class balancing, classifier-specific hyperparameter optimisation, model fitting, and performance evaluation.

An overview of the complete computational study design is presented in Figure 1.

Three sequential feature-selection strategies were evaluated. Each strategy was tested with and without the Synthetic Minority Over-sampling Technique (SMOTE) and was combined with ten machine-learning classifiers. Therefore, each dataset configuration included

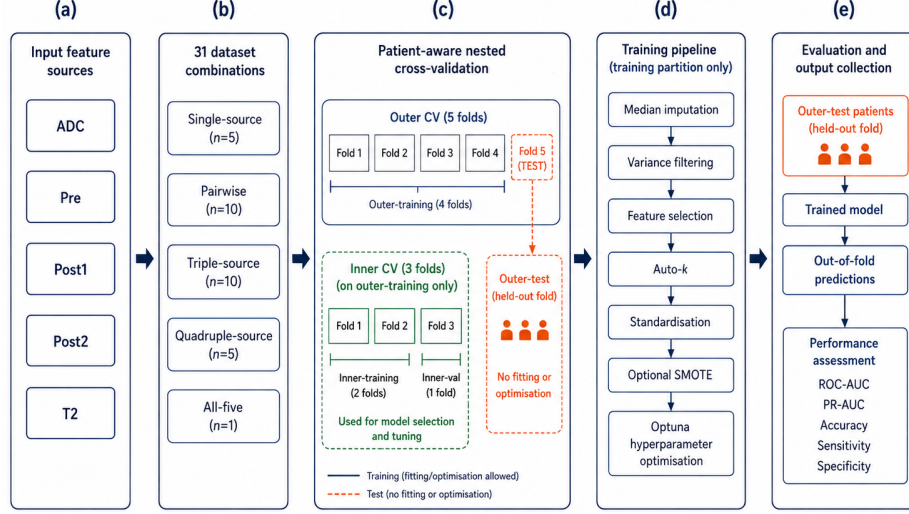


Figure 1: **Overview of the computational machine-learning workflow.** (a) Five pre-extracted radiomic feature sources were used as model inputs: ADC, Pre, Post1, Post2, and T2. (b) The sources were evaluated as five single-source, ten pairwise, ten triple-source, five quadruple-source, and one complete five-source configuration. (c) Patients were separated using five-fold outer and three-fold inner patient-aware nested cross-validation. (d) Median imputation, variance filtering, feature selection, automatic feature-count selection, standardisation, optional SMOTE, and hyperparameter optimisation were restricted to the relevant training partitions. (e) The fitted models were evaluated on outer-test patients, and out-of-fold predictions were collected for performance assessment. ADC, apparent diffusion coefficient; ROI, region of interest; SMOTE, Synthetic Minority Over-sampling Technique.

used only for final performance evaluation.

$$3 \times 2 \times 10 = 60 \quad (2)$$

candidate modelling pipelines. Across the 31 dataset configurations, this produced 1,860 primary combinations of dataset, feature-selection strategy, class-balancing condition, and classifier. Cross-validation folds and hyperparameter-optimisation trials were repeated internal steps and were not counted as separate experimental configurations.

To reduce the risk of information leakage, all ROIs belonging to the same patient were assigned to the same cross-validation partition. Furthermore, all data-dependent operations, including imputation, variance filtering, standardisation, feature selection, SMOTE, and hyperparameter optimisation, were performed using the relevant training data only. The corresponding test data were

2.2 Input Datasets and Prediction Task

The machine-learning analysis used tabular comma-separated value (CSV) files containing pre-extracted numerical radiomic features. The original medical images were not provided directly to the machine-learning models. Instead, each model received a table of numerical measurements representing the corresponding regions of interest.

Each row in an input dataset represented one region of interest (ROI), and each numerical feature column represented one candidate predictor. In addition to the numerical features, each row contained a patient identifier, an ROI identifier, and a

binary class label.

The patient identifier was originally stored in the `INFO_PatientName` column and was renamed to `PatientID` during data preparation. The ROI identifier was stored in `INFO_NameOfRoi`, and the prediction target was stored in `Label`. These meta-data columns were used for data matching, quality control, and patient-aware data splitting, but they were excluded from the predictive feature matrix.

The target labels were encoded as

$$y_i = \begin{cases} 0, & \text{benign,} \\ 1, & \text{malignant,} \end{cases} \quad (3)$$

where y_i denotes the known class label of ROI i . The objective of the classifiers was to estimate whether a previously unseen ROI belonged to the benign or malignant class.

For a dataset containing n ROIs and p numerical predictor features, the model input was represented by

$$\mathbf{X} \in \mathbb{R}^{n \times p}, \quad \mathbf{y} = [y_1 \ y_2 \ \cdots \ y_n]^T, \quad (4)$$

$$\mathbf{g} = [g_1 \ g_2 \ \cdots \ g_n]^T.$$

where $\mathbf{X}$ is the numerical feature matrix, $\mathbf{y}$ is the class-label vector, and g_i identifies the patient associated with ROI i . Each row of $\mathbf{X}$ represented one ROI, whereas each column represented one numerical radiomic feature.

The ROI was the prediction unit because a separate class prediction was generated for each ROI. However, the patient was the grouping unit during cross-validation. Some patients contributed more than one ROI, and these observations could not be considered fully independent. Therefore, all ROIs belonging to the same patient were assigned to the same training, validation, or test partition.

This design prevented a model from being trained on one ROI from a patient and evaluated on another ROI from the same patient. Although predictions and performance metrics were calculated at the ROI level, each outer test fold contained patients who were not used during the corresponding model-development stage.

Five single-source datasets were available: ADC, Pre, Post1, Post2, and T2. The input

data represented 48 unique patients, although the number of available ROI records differed slightly among sources. The single-source datasets contained between 114 and 117 ROIs and between 144 and 145 numerical predictor features.

Table 1 summarises the single-source datasets used to construct the experimental configurations.

The class counts were not identical, but the difference between the two classes was limited. Therefore, class balancing was not applied automatically. Models with and without SMOTE were evaluated as separate experimental conditions, as described in Section 2.8.

2.3 Construction of Multi-Source Datasets

Multi-source datasets were created by combining the feature columns of corresponding ROIs from different sources. The datasets were concatenated column-wise rather than stacked row-wise. Therefore, combining multiple sources increased the number of candidate predictor features but did not create additional patients or ROIs.

For a combination of m feature sources,

$$\mathcal{S} = \{s_1, s_2, \dots, s_m\}, \quad (5)$$

the combined feature matrix was defined as

$$\mathbf{X}^{(\mathcal{S})} = [\mathbf{X}_{\mathcal{I}_{\mathcal{S}}}^{(s_1)} \mid \mathbf{X}_{\mathcal{I}_{\mathcal{S}}}^{(s_2)} \mid \cdots \mid \mathbf{X}_{\mathcal{I}_{\mathcal{S}}}^{(s_m)}], \quad (6)$$

where $\mathcal{I}_{\mathcal{S}}$ represents the set of ROIs that were available in every source included in the combination. Only these shared ROIs were retained. Consequently, the number of observations could differ slightly among source combinations.

Before dataset matching, fully empty columns and exported index columns beginning with `Unnamed:` were removed. The target and patient-group columns were standardised to `Label` and `PatientID`, respectively. A short source tag, such as `ADC`, `Pre`, `Post1`, `Post2`, or `T2`, was assigned to each input dataset.

The corresponding observations were primarily matched using a shared ROI identifier. The `INFO_NameOfRoi` column was preferred when it

Table 1: Characteristics of the five single-source datasets used in the machine-learning analysis. Class 0 represents benign regions of interest, and class 1 represents malignant regions of interest. Metadata and non-predictive identifier columns were not included in the reported feature counts.

Feature source	ROIs, n	Predictor features, n	Class 0, n	Class 1, n
ADC	114	144	50	64
Pre	117	145	52	65
Post1	116	145	52	64
Post2	117	145	52	65
T2	117	145	52	65

was available and unique in every source included in the combination. When required, a combined key containing PatientID and the ROI identifier was used.

A matching key was accepted only when it contained no duplicated values within any of the selected sources. This requirement prevented many-to-many joins, in which a single observation could otherwise generate multiple unintended rows after merging.

When no suitable unique identifier was available, row-order alignment was permitted only if the datasets contained the same number of observations and the complete sequence of patient identifiers was identical. This fallback was used only after the patient ordering had been verified.

Matched observations were combined using a one-to-one inner join. An ROI was therefore retained only when its matching identifier was present in every source included in the combination. Dataset creation was stopped when the selected key was not unique or when no matched observations were found.

Because the five sources contained features with the same original names, every predictive feature was prefixed with its source tag before concatenation. For example,

ADC__GLCM_Correlation
Post1__GLCM_Correlation
T2__GLCM_Correlation

were retained as three separate predictors. This naming procedure prevented duplicated column names and preserved the origin of every feature.

Patient identifiers, ROI identifiers, and class labels were retained as shared metadata and were not given source-specific prefixes.

Each source contained its own copy of the patient identifier and class label. These values were compared after matching. For every retained ROI i , the following consistency conditions were required:

$$\begin{aligned} y_i^{(s_1)} &= y_i^{(s_2)} = \dots = y_i^{(s_m)}, \\ g_i^{(s_1)} &= g_i^{(s_2)} = \dots = g_i^{(s_m)}. \end{aligned} \quad (7)$$

where $y_i^{(s)}$ and $g_i^{(s)}$ denote the class label and patient identifier of ROI i in source s , respectively.

If a class label or patient identifier differed among the matched sources, the affected observations were written to a mismatch report, and creation of that dataset was stopped. After consistency had been confirmed, only one Label column and one PatientID column were retained.

The same construction procedure was applied to all ten pairwise, ten triple-source, five quadruple-source, and one complete five-source combination. Each final dataset contained the shared metadata, all source-prefixed numerical features, and one binary target column.

A dataset-creation log was generated for every attempted combination. The log recorded the included sources, matching key, matching method, number of input and retained ROIs, number of features contributed by each source, class distribution, execution status, and any detected error. These validated datasets were subsequently analysed using the same machine-learning workflow.

2.4 Data Preparation and Quality Control

Before model development, all input datasets underwent the same data-preparation and quality-control procedure. The purpose of this stage was to confirm that the datasets had the expected structure, identify invalid or duplicated records, and define the numerical predictor columns. No model was trained during this stage.

Each input file was checked for the required patient identifier, ROI identifier, and binary class-label columns. The original patient identifier, `INFO_PatientName`, was renamed to `PatientID`. The ROI identifier was retained as `INFO_NameOfRoi`, and the target variable was retained as `Label`.

The target column was required to contain only the values 0 and 1. Rows with missing patient identifiers or missing class labels were not eligible for patient-aware model development. The patient identifier, ROI identifier, target label, exported index columns, and other non-predictive metadata were excluded from the model input.

All remaining numerical columns were treated as candidate predictors. Columns containing no valid numerical values were removed. Rows with partially missing predictor values were retained because missing-value imputation was later performed separately within each training partition.

Feature names were cleaned to improve consistency and compatibility with the analysis code. Leading and trailing spaces, parenthetical text, and unsupported characters were removed, and spaces were replaced with underscores. Long feature-family names were shortened using the following replacements:

- MORPHOLOGICAL was replaced by Morph;
- INTENSITY-BASED was replaced by IB;
- INTENSITY-HISTOGRAM was replaced by IH.

If two columns produced the same cleaned name, a numerical suffix was added to preserve unique column names. A mapping between the

original and cleaned names was saved to maintain traceability.

The number of missing and infinite values was recorded for each candidate feature. Missing values were not replaced using statistics calculated from the complete dataset. Instead, median imputation was performed later using only the corresponding training data within cross-validation. Infinite values were treated as invalid numerical entries and were included in the data-quality report.

Feature variance was also examined. Features with variance less than or equal to 10^{-12} were classified as constant, whereas features with variance greater than 10^{-12} but less than or equal to 10^{-8} were classified as near-zero-variance features. These values were reported during quality control. The variance filter used during model development was subsequently fitted again within each training partition.

The workflow checked for fully duplicated rows, repeated ROI identifiers, duplicated patient-ROI pairs, and repeated feature names. Suspected duplicated observations were reported rather than removed automatically because their exclusion required confirmation that they represented repeated exports rather than valid separate records.

The number of ROIs contributed by each patient was also recorded. Patients with more than one ROI were retained. Patients who had ROIs from both target classes were also retained because classification was performed at the ROI level. However, all ROIs from the same patient were later assigned to the same cross-validation partition.

Potential outlying values were assessed using the modified z -score,

$$z_{ij}^* = 0.6745 \frac{x_{ij} - \text{median}(x_j)}{\text{MAD}(x_j)}, \quad (8)$$

where x_{ij} is the value of feature j for ROI i , and $\text{MAD}(x_j)$ is the median absolute deviation of feature j . Values satisfying

$$|z_{ij}^*| > 3.5 \quad (9)$$

were flagged as potential outliers.

Outlier detection was used only as a diagnostic procedure. No ROI was removed, and no global

clipping or winsorisation was applied in the main experiments. This avoided estimating transformation limits from the complete dataset before cross-validation.

The class distribution was recorded for every dataset configuration. Because the difference between the numbers of benign and malignant ROIs was limited, class balancing was not applied as a mandatory preprocessing step. Instead, pipelines with and without SMOTE were evaluated separately.

The quality-control stage produced structured summaries of feature names, missing and infinite values, feature variance, duplicated records, patient-level ROI counts, class distributions, mixed-label patients, and potential outliers. These outputs provided a reproducible record of the data supplied to the machine-learning pipeline.

No global imputation, standardisation, supervised feature selection, SMOTE, hyperparameter optimisation, or classifier fitting was performed during this stage. All such operations were restricted to the relevant training partitions during patient-aware nested cross-validation.

2.5 Patient-Aware Nested Cross-Validation

Model development and internal performance evaluation were performed using patient-aware nested cross-validation. This design was used to separate model optimisation from final performance estimation and to account for patients who contributed more than one ROI.

Nested cross-validation contains two consecutive validation loops. The outer loop estimates how well the completed modelling procedure performs on previously unseen patients. The inner loop is used only for model development, including selection of the feature-set size and optimisation of classifier hyperparameters.

2.5.1 Patient-aware data splitting

Some patients contributed multiple ROIs. Therefore, individual rows could not be divided independently between the training and test sets. If

ROIs from the same patient appeared in both partitions, the model could be trained using patient-specific information that was also present in the test data.

The patient identifier was consequently used as the grouping variable. All ROIs belonging to one patient were assigned to the same cross-validation partition. For outer fold q , the patient groups in the training and test sets were required to satisfy

$$\mathcal{G}_{\text{train}}^{(q)} \cap \mathcal{G}_{\text{test}}^{(q)} = \emptyset, \quad (10)$$

where $\mathcal{G}_{\text{train}}^{(q)}$ and $\mathcal{G}_{\text{test}}^{(q)}$ denote the sets of patients in the corresponding outer-training and outer-test partitions.

The absence of overlapping patient identifiers was verified programmatically in every outer fold. Model execution was stopped if a patient was detected in both partitions.

The folds were generated using `StratifiedGroupKFold`. This method kept all observations from one patient together while preserving the distribution of benign and malignant ROIs across folds as closely as possible. Because patients contributed different numbers of ROIs, identical class proportions could not always be achieved.

The observations were shuffled before fold construction, and a fixed random seed of 100 was used to improve reproducibility.

2.5.2 Outer cross-validation loop

The outer loop contained five patient-aware folds:

$$K_{\text{outer}} = 5. \quad (11)$$

In each iteration, approximately four-fifths of the patient groups were used for model development, and the remaining patients formed the outer-test set. The exact numbers of ROIs could differ among folds because splitting was performed at the patient level.

The outer-test partition remained unavailable throughout model development. It was not used to:

- calculate missing-value replacement values;

- identify low-variance features;
- select the number or identity of retained features;
- calculate standardisation parameters;
- generate synthetic SMOTE observations;
- optimise classifier hyperparameters.

After all model-development decisions had been completed, the selected pipeline was fitted using the complete outer-training partition and evaluated once on the corresponding outer-test partition.

2.5.3 Training-only preprocessing

Preprocessing parameters were estimated separately within each outer-training partition. Missing feature values were first replaced using the median of the corresponding feature in the outer-training data. For feature j in outer fold q , the fitted median was

$$m_j^{(q)} = \text{median} \{x_{ij} : i \in \mathcal{D}_{\text{train}}^{(q)}\}. \quad (12)$$

The same fitted value was used to replace missing entries in both the outer-training and outer-test partitions. The outer-test median was not calculated separately.

After imputation, a variance filter was fitted using the outer-training features. Features with variance less than or equal to 10^{-10} were removed. The same fitted feature mask was then applied to the outer-test data.

After feature selection, the retained features were standardised using the mean and standard deviation calculated from the training partition:

$$z_{ij} = \frac{x_{ij} - \mu_{j,\text{train}}}{\sigma_{j,\text{train}}}, \quad (13)$$

where $\mu_{j,\text{train}}$ and $\sigma_{j,\text{train}}$ represent the training-derived mean and standard deviation of feature j . The same values were applied unchanged to the corresponding validation or test data.

This procedure ensured that the outer-test observations followed the transformations learned from

the training data but did not influence those transformations.

2.5.4 Inner cross-validation loop

The current outer-training partition was further divided using a three-fold patient-aware inner loop:

$$K_{\text{inner}} = 3. \quad (14)$$

The inner loop was used to compare candidate feature counts and classifier hyperparameters. Patient grouping was maintained so that the inner-training and inner-validation patient sets were also mutually exclusive.

When the number of available patient groups or class observations did not support three valid inner folds, the implementation reduced the number of inner folds to the largest feasible value. At least two valid inner folds were required.

For Auto- k selection, the inner folds were generated from the outer-training feature matrix after outer-fold median imputation and variance filtering. Within each inner split, the active feature-selection procedure and standardisation were fitted using the inner-training data and applied to the corresponding inner-validation data. When SMOTE was enabled, it was applied only to the transformed inner-training observations. The inner-validation samples retained their original values and class distribution.

ROC-AUC on the inner-validation folds was used to compare candidate settings. The selected settings were then used to reconstruct the classifier from the complete outer-training data.

2.5.5 Final outer-fold evaluation and out-of-fold predictions

After the inner model-development process had finished, the selected feature-selection procedure and classifier settings were applied to the complete outer-training partition. The fitted classifier then generated continuous prediction scores and class labels for the outer-test observations.

Each ROI appeared in an outer-test fold once during the five-fold procedure. Its prediction was therefore produced by a model that had not been

trained using that ROI or any other ROI from the same patient.

Predictions from the five outer-test folds were combined to form the out-of-fold prediction vector

$$\hat{\mathbf{p}}_{\text{OOF}} = [\hat{p}_1, \hat{p}_2, \dots, \hat{p}_n]^T. \quad (15)$$

These predictions were subsequently used to calculate pooled performance metrics and construct ROC curves, precision–recall curves, confusion matrices, and calibration summaries. Fold-specific results were also retained to describe variation in performance across different outer-test patient groups.

2.6 Feature-Selection Strategies

Feature selection was used to reduce the number of input variables before classifier training. This step was particularly important for the multi-source datasets, in which the number of candidate features increased while the number of independent patients remained almost unchanged.

A compact feature set reduces the computational complexity of model training and limits the number of variables from which the classifier can learn. This can reduce the risk of overfitting in datasets where the number of candidate predictors is much larger than the number of patients.

Three sequential feature-selection strategies were compared:

1. correlation filtering followed by minimum redundancy maximum relevance (mRMR);
2. Elastic Net followed by mRMR;
3. mRMR followed by Elastic Net, without an initial correlation-filtering step.

The order of the feature-selection operations was treated as an experimental factor. The first selector determined which variables were available to the second selector. Consequently, Elastic Net followed by mRMR and mRMR followed by Elastic Net were evaluated as two different pipelines rather than as equivalent procedures.

All feature-selection steps were fitted using the relevant training partition only. The resulting feature indices were subsequently applied unchanged to the corresponding validation or test data. SMOTE was not used during feature selection and was applied only after the final feature subset had been selected and standardised.

2.6.1 Correlation Filtering

Correlation filtering was used to remove strongly redundant variables before mRMR selection. For each training partition, pairwise Pearson correlations were calculated between the candidate features.

Two features were considered strongly correlated when

$$|r_{jk}| > 0.90, \quad (16)$$

where r_{jk} represents the Pearson correlation between features j and k .

When several features were strongly correlated, the workflow did not remove one of them arbitrarily. Each feature was first evaluated using its univariate ROC-AUC. Because a feature could be predictive in either direction, a direction-independent score was calculated as

$$A_j^* = \max(A_j, 1 - A_j), \quad (17)$$

where A_j is the ROC-AUC obtained using feature j alone.

Features were processed from the highest to the lowest value of A_j^* . The highest-ranked remaining feature was retained, and other remaining features with an absolute correlation greater than 0.90 with that feature were removed. This process was repeated until all candidate features had been considered.

The correlation matrix and univariate ROC-AUC values were calculated separately within each training partition. Therefore, the validation and test data did not influence which correlated features were retained.

2.6.2 Minimum Redundancy Maximum Relevance

Minimum redundancy maximum relevance was used to identify features that were informative about the class label while avoiding repeated information in the selected subset.

The method considered two properties:

- *relevance*, describing how much information a feature contained about the target class;
- *redundancy*, describing how similar a candidate feature was to features that had already been selected.

The relevance of candidate feature j was estimated using mutual information:

$$R_j = I(X_j; Y), \quad (18)$$

where X_j denotes feature j , Y denotes the binary class label, and $I(\cdot; \cdot)$ represents mutual information.

Mutual information can identify both linear and nonlinear relationships between a feature and the target. It was estimated using `mutual_info_classif`. The number of neighbours used by this estimator was adjusted to the number of minority-class observations in the current training partition:

$$n_{\text{MI}} = \max [1, \min (3, n_{\text{minority}} - 1)]. \quad (19)$$

This adjustment prevented the estimator from requesting more neighbours than were supported by a small training fold.

For a candidate feature j , redundancy was calculated as the mean absolute Pearson correlation with the features already selected:

$$D_j = \frac{1}{|\mathcal{F}|} \sum_{s \in \mathcal{F}} |r_{js}|, \quad (20)$$

where $\mathcal{F}$ is the current selected feature set. The mRMR score was then calculated as

$$Q_j = R_j - D_j. \quad (21)$$

The first feature was selected using relevance alone because no previously selected feature was available for calculating redundancy. Subsequent variables were added one at a time by selecting the remaining feature with the largest value of Q_j . This process continued until the required number of features had been selected.

2.6.3 Elastic-Net Feature Selection

Elastic-Net feature selection was implemented using regularised logistic regression. Elastic Net combines two forms of regularisation: the L_1 penalty, which encourages some feature coefficients to become small or zero, and the L_2 penalty, which improves stability when several predictors contain related information.

The Elastic-Net objective can be expressed as

$$\mathcal{L}_{\text{EN}} = \mathcal{L}_{\text{logistic}} + \lambda \left[\alpha \sum_{j=1}^p |\beta_j| + \frac{1-\alpha}{2} \sum_{j=1}^p \beta_j^2 \right], \quad (22)$$

where β_j is the fitted coefficient of feature j , λ controls the overall strength of regularisation, and α controls the balance between the L_1 and L_2 components.

The candidate features were standardised before fitting the Elastic-Net selector. The logistic-regression model used an `l1_ratio` of 0.8 and a regularisation parameter C of 0.1. It was fitted using the `saga` solver, balanced class weights, a maximum of 5000 iterations, a convergence tolerance of 10^{-5} , and a random seed of 100.

After fitting, features were ranked according to the absolute magnitude of their coefficients:

$$W_j = |\beta_j|. \quad (23)$$

The highest-ranked features were retained. Absolute coefficient values were used because both positive and negative coefficients could contribute to separation of the two classes.

2.6.4 Sequential Feature-Selection Pipelines

The three component methods were combined into the following sequential pipelines.

Correlation filtering followed by mRMR. Correlation filtering was first applied to the complete training feature matrix. The purpose of this first stage was to remove strongly correlated variables while retaining the member of each correlated group with the highest direction-independent univariate ROC-AUC.

The final feature subset was then selected from the reduced candidate space using mRMR:

$$\begin{aligned} \text{Correlation filtering} &\rightarrow \text{mRMR} \\ &\rightarrow \text{final subset.} \end{aligned} \quad (24)$$

The number of variables retained by the final mRMR stage was determined by the Auto- k procedure described in Section 2.7.

Elastic Net followed by mRMR. Elastic Net was first fitted to the complete training feature space and used to retain a candidate pool of up to 28 variables. If fewer than 28 features were available, all available variables were considered.

The final subset was then selected from this candidate pool using mRMR:

$$\begin{aligned} \text{Elastic Net} &\rightarrow \text{candidate pool} \\ &\rightarrow \text{mRMR} \rightarrow \text{final subset.} \end{aligned} \quad (25)$$

The purpose of this order was to perform an initial supervised and regularised reduction of the feature space, followed by selection of a smaller subset with reduced redundancy.

mRMR followed by Elastic Net. The third strategy reversed the order of the previous pipeline. mRMR was first applied to the complete training feature matrix and retained a candidate pool of up to 28 features. Elastic Net was then fitted to this reduced representation and ranked the candidate variables using their absolute coefficients:

$$\begin{aligned} \text{mRMR} &\rightarrow \text{candidate pool} \\ &\rightarrow \text{Elastic Net} \rightarrow \text{final subset.} \end{aligned} \quad (26)$$

No separate correlation filter was applied before mRMR in this pipeline. Redundancy was addressed directly by the initial mRMR selection stage.

Although this strategy used the same two component methods as the Elastic-Net-mRMR pipeline, the two procedures were not interchangeable. The first stage produced a different candidate space for the second stage and could therefore lead to a different final feature subset.

For every outer fold, the active feature-selection pipeline was fitted using the available outer-training data after median imputation and variance filtering. During Auto- k selection, the same procedure was repeated independently within the patient-aware inner folds.

After the final feature count had been selected, the feature-selection pipeline was refitted using the complete outer-training partition. The selected feature indices were then applied unchanged to the corresponding outer-test data.

Within a given dataset, feature-selection strategy, SMOTE condition, and outer fold, the same selected feature subset was supplied to all ten classifiers. This enabled the classifiers to be compared using an identical input representation.

The selected feature names were saved separately for every outer fold. These records were later used to calculate feature-selection frequencies and assess the stability of the resulting feature subsets.

2.7 Automatic Selection of the Feature-Set Size

The final number of retained features was selected automatically within each outer cross-validation fold. This procedure, referred to as Auto- k , allowed model complexity to adapt to the patients available in the current outer-training partition rather than using one fixed feature count for all datasets and folds.

The candidate feature counts were

$$\mathcal{K} = \{4, 6, 8, 10, 12, 14, 16, 18\}. \quad (27)$$

Candidate values larger than the number of features remaining after outer-fold imputation and variance filtering were excluded. The set of valid candidates could therefore differ if a training partition contained fewer usable variables.

Auto- k was performed independently for each outer fold and feature-selection strategy. Only the current outer-training data were used. The corresponding outer-test data were not used to determine the number or identity of the retained features.

The outer-training data were divided into up to three patient-aware inner folds using `StratifiedGroupKFold`. All ROIs from the same patient remained in the same inner partition. The number of inner folds was restricted by the requested fold count, the size of the smaller class, and the number of available patient groups. At least two valid inner folds were required.

For each candidate value k , the active feature-selection pipeline was fitted separately using the inner-training partition and was requested to retain k variables. The selected feature indices were then applied unchanged to the corresponding inner-validation partition.

The selected features were standardised using parameters estimated from the inner-training data. For feature j , the same training-derived mean and standard deviation were used to transform the inner-validation observations.

A lightweight Logistic Regression classifier was used only to compare the candidate feature counts. This tuning classifier used L_2 regularisation, the `liblinear` solver, balanced class weights, a maximum of 5000 iterations, and a random seed of 100. It was not necessarily the final classifier evaluated in the outer fold.

For candidate k , the ROC-AUC values obtained from the valid inner folds were averaged:

$$S_k = \frac{1}{R_k} \sum_{r \in \mathcal{R}_k} \text{AUC}_{k,r}, \quad (28)$$

where $\mathcal{R}_k$ is the set of inner folds with a valid

ROC-AUC, and R_k is the number of valid folds. An inner-validation fold containing only one class was not assigned a ROC-AUC value.

The selected feature count was

$$k^* = \arg \max_{k \in \mathcal{K}} S_k. \quad (29)$$

When two candidate feature counts produced the same mean inner-fold ROC-AUC, the smaller value of k was selected. This rule favoured the more compact feature representation when predictive performance was equal.

SMOTE was not applied during Auto- k selection. Class imbalance was instead addressed by the balanced class weights of the tuning classifier. This separated selection of the feature-set size from the later comparison of SMOTE and no-SMOTE pipelines.

After k^* had been selected, the active feature-selection pipeline was refitted using the complete outer-training feature matrix. The resulting feature subset was applied unchanged to the corresponding outer-test data.

Within each outer fold, the same Auto- k -selected feature subset was provided to all ten classifiers evaluated under the corresponding feature-selection and SMOTE condition. The feature-count search was therefore performed once per outer fold and experimental run rather than separately for every classifier.

For reproducibility, the candidate values, individual inner-fold ROC-AUC scores, mean and standard deviation of the scores, tuning classifier, tuning metric, and selected value of k were saved for each outer fold.

2.8 Class-Imbalance Handling Using SMOTE

The numbers of benign and malignant ROIs were not identical, although the observed class imbalance was limited. Class balancing was therefore treated as an experimental factor rather than as a mandatory preprocessing operation.

Each feature-selection strategy was evaluated under two conditions:

1. without synthetic oversampling;
2. with the Synthetic Minority Over-sampling Technique (SMOTE).

This comparison was performed separately for every dataset configuration and classifier. It allowed the effect of SMOTE to be evaluated rather than assuming that oversampling would improve all models.

2.8.1 Position Within the Modelling Pipeline

SMOTE identifies neighbouring observations using distances in the feature space. It was therefore applied after feature selection and standardisation, when the retained variables had comparable numerical scales.

For the final model fitted in each outer fold, the main processing sequence was

$$\begin{aligned}
 &\text{Imputation} \rightarrow \text{variance filtering} \\
 &\quad \rightarrow \text{feature selection} \\
 &\quad \rightarrow \text{standardisation} \\
 &\quad \rightarrow \text{optional SMOTE} \\
 &\quad \rightarrow \text{classifier fitting.}
 \end{aligned} \tag{30}$$

Feature selection was performed before SMOTE so that synthetic samples did not influence the number or identity of the selected variables. Similarly, SMOTE was not used during the Auto- k procedure described in Section 2.7.

2.8.2 Training-Only Application

SMOTE was applied only after the training and evaluation partitions had been separated. It was never fitted to the complete dataset, inner-validation data, or outer-test data.

During classifier-specific hyperparameter optimisation, preprocessing was fitted separately within each inner split. Missing values were imputed using the inner-training medians, low-variance features were identified from the inner-training data, and standardisation was fitted to the inner-training features. When SMOTE was enabled, it was then applied only to the transformed inner-training observations.

The candidate classifier was evaluated on the unchanged inner-validation observations. These validation samples retained their original class distribution and were not used to generate synthetic training data.

After hyperparameter selection, the selected features were standardised using the complete outer-training partition. SMOTE was then applied to the standardised outer-training data before fitting the final classifier for that fold. The corresponding outer-test data were transformed using the training-derived parameters but were not over-sampled.

Consequently, every reported validation or test prediction was generated for an original ROI that had not been used during synthetic sample generation.

2.8.3 Adaptive Neighbour Selection

The standard implementation of SMOTE commonly uses five nearest minority-class neighbours. Some inner-training partitions, however, contained too few minority-class observations to support this setting.

The number of neighbours was therefore selected adaptively as

$$k_{\text{SMOTE}} = \min(5, n_{\text{minority}} - 1), \tag{31}$$

where n_{minority} is the number of minority-class observations in the current training partition.

SMOTE was not applied when the training partition contained only one class or fewer than two minority-class observations. In these cases, the original training data were retained.

The original training data were also retained if resampling could not be completed because of an unexpected numerical or structural error. A fixed random seed of 100 was used for reproducibility.

2.8.4 Experimental Comparison

The three feature-selection strategies were evaluated both with and without SMOTE. Each balancing condition was combined with the same ten classifiers. This produced

$$3 \times 2 \times 10 = 60 \quad (32)$$

candidate modelling pipelines for each dataset configuration.

The two balancing conditions used the same patient-aware outer folds, feature-selection strategy, candidate feature counts, classifiers, and evaluation metrics. Their results were compared using mean and pooled ROC-AUC, PR-AUC, balanced accuracy, sensitivity, specificity, precision, and F1-score.

SMOTE was therefore evaluated as a classifier-training condition. It was not considered part of the original dataset and did not alter the outer-test observations or their class distribution.

2.9 Machine-Learning Classifiers

Ten supervised machine-learning classifiers were evaluated to compare different approaches to binary classification. The selected models included linear, kernel-based, tree-based, distance-based, and probabilistic methods.

Each classifier received the feature subset selected within the corresponding outer cross-validation fold. Patient identifiers, ROI identifiers, class labels, and other metadata columns were excluded from the model input.

All selected numerical features were standardised before classifier fitting. Although tree-based models are generally less sensitive to feature scale, the same standardised representation was supplied to all classifiers to maintain a consistent modelling workflow.

For ROI i , a fitted classifier generated a continuous estimate of the positive-class output:

$$\hat{p}_i = f_{\theta}(\mathbf{x}_i), \quad (33)$$

where $\mathbf{x}_i$ contains the selected features for ROI i , and θ denotes the fitted model parameters. Whenever available, the predicted probability of the malignant class was retained. Continuous decision scores were used when a probability output was not available.

The continuous outputs were used to calculate threshold-independent metrics, including ROC-

AUC and PR-AUC. Class labels were subsequently obtained using the predefined classification threshold described in Section 2.11.

2.9.1 Linear and Kernel-Based Classifiers

Logistic Regression was used as a regularised linear classifier. It models the probability of the malignant class as a function of a weighted combination of the selected features. The implemented model used Elastic-Net regularisation, which combines L_1 and L_2 penalties. The L_1 component encourages a more compact coefficient structure, whereas the L_2 component improves stability when the input variables contain related information.

The Logistic Regression classifier used the saga solver, balanced class weights, a maximum of 5000 iterations, and a convergence tolerance of 10^{-4} . The regularisation parameter C and the relative contribution of the L_1 penalty were selected within the inner cross-validation loop.

Two support vector machine classifiers were also evaluated. A support vector machine identifies a decision boundary that separates the two classes while maintaining the largest possible margin between them.

The linear SVM used a linear decision boundary and was intended to evaluate whether the selected features could be separated using a relatively simple model. Its regularisation parameter C was optimised within the inner loop.

The RBF SVM used a radial basis function kernel to represent nonlinear relationships between the selected features and the class label. Its regularisation parameter C and kernel-width parameter γ were optimised.

Both SVM classifiers used balanced class weights and enabled probability estimation. Probability estimation allowed their outputs to be evaluated using the same probability-based metrics as the other classifiers.

2.9.2 Tree-Based Ensemble Classifiers

Five tree-based ensemble classifiers were evaluated: Random Forest, Extra Trees, Gradient

Boosting, XGBoost, and LightGBM. These methods combine multiple decision trees rather than relying on one individual tree.

Random Forest constructs several decision trees using random subsets of training observations and predictor features. The predictions from the individual trees are then combined. This approach can model nonlinear relationships and interactions among features while reducing the instability associated with a single decision tree.

The number of trees, maximum tree depth, minimum number of observations per terminal leaf, and number of candidate features considered at each split were optimised. Balanced class weights were used during model fitting.

Extra Trees is also an ensemble of decision trees but introduces additional randomness when selecting split thresholds. This increased randomisation can reduce correlation among the individual trees. The number of trees, maximum depth, minimum leaf size, and feature-sampling strategy were optimised. Balanced class weights were used.

Gradient Boosting constructs decision trees sequentially. Each new tree is trained to reduce the prediction errors remaining from the previous trees. The number of trees, maximum tree depth, learning rate, and fraction of training observations used by each boosting stage were optimised.

XGBoost is a regularised gradient-boosting implementation. It sequentially constructs trees while controlling model complexity through feature subsampling, observation subsampling, and explicit L_1 and L_2 regularisation. The tree number and depth, learning rate, subsampling parameters, and regularisation parameters were optimised.

LightGBM is another gradient-boosted decision-tree implementation designed for computationally efficient model construction. The evaluated parameters included the number and depth of trees, number of terminal leaves, learning rate, feature and observation subsampling, and L_1 and L_2 regularisation. Balanced class weights were used during fitting.

2.9.3 Distance-Based and Probabilistic Classifiers

K-Nearest Neighbours (KNN) was included as a distance-based classifier. For a new ROI, KNN identifies the most similar training observations in the standardised feature space and assigns a class according to their labels.

The number of neighbouring observations, the use of uniform or distance-based voting, and the distance metric were optimised. Euclidean and Manhattan distances were considered. Standardisation was particularly important for KNN because features with larger numerical ranges would otherwise dominate the distance calculation.

Gaussian Naive Bayes was included as a probabilistic classifier. It models the distribution of each selected feature separately within the benign and malignant classes and combines these distributions to estimate the class probability.

The method assumes that the selected features follow approximately Gaussian class-conditional distributions and are conditionally independent after the class is known. Although this assumption may not hold exactly for radiomic data, the model provides a computationally simple probabilistic reference. Its variance-smoothing parameter was optimised to reduce numerical instability when a feature had a small within-class variance.

Within every experimental run, the ten classifiers were evaluated using the same outer patient folds, selected feature subset, SMOTE condition, and performance measures. Classifier-specific hyperparameters were selected independently within the inner cross-validation loop.

The outer-test predictions were generated only after hyperparameter selection had been completed. Consequently, the comparison among classifiers was based on observations from patients who were not used for classifier fitting or optimisation.

2.10 Hyperparameter Optimisation Using Optuna

The behaviour of a machine-learning classifier depends partly on its hyperparameters. Hyperparam-

Table 2: Machine-learning classifiers evaluated in the computational framework. Hyperparameters were selected independently within each outer fold using patient-aware inner cross-validation.

Classifier		Model family	Optimised hyperparameters
Logistic Regression		Regularised linear model	Regularisation strength C and Elastic-Net <code>l1_ratio</code> .
Linear SVM		Linear margin-based model	Regularisation strength C .
RBFSVM		Nonlinear kernel model	Regularisation strength C and kernel parameter γ .
Random Forest		Bagged tree ensemble	Number of trees, maximum depth, minimum leaf size, and feature-sampling strategy.
Extra Trees		Randomised tree ensemble	Number of trees, maximum depth, minimum leaf size, and feature-sampling strategy.
Gradient Boosting		Sequential tree ensemble	Number of trees, maximum depth, learning rate, and observation subsampling.
K-Nearest Neighbours		Distance-based model	Number of neighbours, voting weights, and distance metric.
Gaussian Naive Bayes		Probabilistic model	Variance-smoothing parameter.
XGBoost		Regularised boosted-tree model	Number and depth of trees, learning rate, observation and feature subsampling, and L_1 and L_2 regularisation.
LightGBM		Gradient-boosted tree model	Number and depth of trees, number of leaves, learning rate, observation and feature subsampling, and L_1 and L_2 regularisation.

eters are model settings that must be chosen before the final model is fitted. Examples include the regularisation strength of a linear model, the number and depth of trees in an ensemble, and the number of neighbours used by K-Nearest Neighbours.

Classifier-specific hyperparameters were optimised using Optuna within the inner loop of the patient-aware nested cross-validation procedure. This ensured that model settings were selected using only the patients available in the current outer-training partition. The corresponding outer-test patients were not used during hyperparameter optimisation.

A separate Optuna study was created for each dataset configuration, feature-selection strategy, SMOTE condition, classifier, and outer fold. Each study evaluated

$$N_{\text{trials}} = 10 \quad (34)$$

candidate hyperparameter configurations. The same trial budget was used for all classifiers to provide a consistent comparison and to control the computational cost of the complete experimental framework.

Candidate configurations were generated using the Tree-structured Parzen Estimator sampler:

```
optuna.samplers.TPESampler(seed=100)
```

The optimisation direction was set to maximise the inner-validation objective. A fixed random seed of 100 was used to make the sampling procedure reproducible.

2.10.1 Inner-Loop Evaluation of Candidate Parameters

Each Optuna trial represented one candidate set of hyperparameters for one classifier. The candidate classifier was evaluated using up to three patient-aware inner folds generated from the current outer-training data.

The actual number of inner folds was restricted by the requested three-fold design, the number of observations in the smaller class, and the number of unique patient groups:

$$K_{\text{inner,actual}} = \min(3, n_{\text{minority}}, n_{\text{patients}}). \quad (35)$$

At least two valid inner folds were required. All ROIs from the same patient remained in the same inner partition.

For every inner split, the following operations were performed:

1. missing values were replaced using medians calculated from the inner-training data;
2. features with a training variance less than or equal to 10^{-10} were removed;
3. standardisation was fitted using the inner-training data and applied unchanged to the inner-validation data;
4. when enabled, SMOTE was applied only to the transformed inner-training observations;
5. the candidate classifier was fitted to the inner-training data;
6. ROC-AUC was calculated using the unchanged inner-validation observations.

When the classifier provided class probabilities, the estimated probability of the malignant class was used to calculate ROC-AUC. Otherwise, a continuous decision-function score was used.

An inner-validation fold containing only one target class was excluded because ROC-AUC could not be calculated for that fold. Model-fitting or prediction failures were assigned an AUC value of 0.5, representing chance-level discrimination.

2.10.2 Optimisation Objective

For trial t , the valid inner-fold ROC-AUC values were combined using their median:

$$\tilde{A}_t = \text{median}_r(A_{t,r}), \quad (36)$$

where $A_{t,r}$ is the validation ROC-AUC obtained by trial t in inner fold r .

The median was used rather than the mean to reduce the influence of an unusually easy or difficult inner-validation fold. The implemented Optuna objective was

$$J_t = \tilde{A}_t - 0.001p, \quad (37)$$

where p is the number of selected features supplied to the classifier.

For a given classifier and outer fold, the selected feature set was fixed before the Optuna study began. Consequently, p was constant across all trials within that study. The complexity term therefore shifted all objective values by the same amount and did not change the ranking of the candidate hyperparameter configurations.

A trial received an objective value corresponding to chance-level performance when fewer than two inner folds were possible, no valid ROC-AUC values remained, all features were removed by variance filtering, or classifier fitting failed.

2.10.3 Hyperparameter Search Spaces

The classifier-specific search spaces were selected to compare plausible model complexities while limiting overfitting and computational cost. Logarithmic sampling was used for parameters that covered several orders of magnitude. The complete search spaces are reported in Table 3.

2.10.4 Selection and Final Outer-Fold Fitting

After all ten trials had been completed, the hyperparameter configuration with the highest inner objective value was selected independently for each classifier and outer fold:

$$\lambda_m^{*(q)} = \arg \max_{\lambda \in \Lambda_m} J_m^{(q)}(\lambda), \quad (38)$$

where m denotes the classifier, q denotes the outer fold, and Λ_m is the classifier-specific search space.

A new classifier was constructed using the selected hyperparameters. It was then fitted using the complete processed outer-training partition. When SMOTE was enabled, the final clas-

sifier was fitted using the resampled outer-training data.

The fitted classifier was evaluated once on the corresponding outer-test partition. The outer-test predictions and performance values did not influence the Optuna study or the selection of the hyperparameters.

The selected hyperparameters, best inner objective value, selected feature count, and outer-test performance measures were saved separately for each classifier and outer fold.

2.11 Model Performance Evaluation and Uncertainty

Model performance was evaluated using predictions generated for the outer-test partitions of the patient-aware nested cross-validation procedure. Each ROI received one out-of-fold prediction from a model that had not been trained using that ROI or any other ROI from the same patient.

The primary model-selection measure was the mean outer-fold receiver operating characteristic area under the curve (ROC-AUC). Additional discrimination, classification, agreement, and calibration measures were calculated to provide a more complete description of model performance.

2.11.1 Primary Performance Measure

ROC-AUC measures how well a classifier ranks malignant ROIs above benign ROIs across all possible classification thresholds. A value of 0.5 represents chance-level ranking, whereas a value of 1 represents complete separation of the two classes.

For each classifier, ROC-AUC was calculated separately in every outer test fold. The primary performance measure was the mean of the five outer-fold values:

$$\overline{\text{AUC}} = \frac{1}{K_{\text{outer}}} \sum_{q=1}^{K_{\text{outer}}} \text{AUC}^{(q)}, \quad (39)$$

where $K_{\text{outer}} = 5$ and $\text{AUC}^{(q)}$ is the ROC-AUC obtained in outer fold q .

The standard deviation of the outer-fold ROC-AUC values was also reported to describe varia-

Table 3: Classifier-specific hyperparameter search spaces used during Optuna optimisation. Log indicates logarithmic sampling. Categorical parameters were selected from the listed alternatives.

Classifier	Hyperparameter	Search space
Logistic Regression	C l1_ratio	10^{-4} to 1, log 0 to 1
Linear SVM	C	10^{-4} to 1, log
RBF SVM	C γ	10^{-3} to 10, log 10^{-4} to 10^{-1} , log
Random Forest	n_estimators max_depth min_samples_leaf max_features	50, 100, 150, or 200 Integer from 2 to 8 Integer from 1 to 10 sqrt, log2, or all available features
Extra Trees	n_estimators max_depth min_samples_leaf max_features	50 to 500 in increments of 50 Integer from 2 to 10 Integer from 2 to 20 sqrt, log2, or all available features
Gradient Boosting	n_estimators max_depth learning_rate subsample	25, 50, 75, or 100 Integer from 1 to 3 0.01 to 0.10, log 0.70 to 1.00
K-Nearest Neighbours	n_neighbors weights metric	Integer from 3 to 15 Uniform or distance-based Euclidean or Manhattan distance
Gaussian Naive Bayes	var_smoothing	10^{-12} to 10^{-6} , log
XGBoost	n_estimators max_depth learning_rate subsample colsample_bytree reg_alpha reg_lambda	50 to 300 in increments of 50 Integer from 2 to 8 0.01 to 0.30, log 0.60 to 1.00 0.50 to 1.00 10^{-8} to 10, log 10^{-8} to 10, log
LightGBM	n_estimators max_depth num_leaves learning_rate subsample colsample_bytree reg_alpha reg_lambda	50 to 300 in increments of 50 Integer from 2 to 8 Integer from 8 to 64 0.01 to 0.30, log 0.60 to 1.00 0.50 to 1.00 10^{-8} to 10, log 10^{-8} to 10, log

tion in performance across different outer-test patient groups. Candidate pipelines were ranked primarily according to mean outer-fold ROC-AUC.

2.11.2 Pooled Out-of-Fold Performance

Predictions from all outer-test folds were combined into one out-of-fold prediction vector. Pooled ROC-AUC was then calculated as

$$\text{AUC}_{\text{pooled}} = \text{AUC}(\mathbf{y}, \hat{\mathbf{p}}_{\text{OOF}}), \quad (40)$$

where $\mathbf{y}$ is the complete class-label vector and $\hat{\mathbf{p}}_{\text{OOF}}$ contains the corresponding out-of-fold predicted probabilities.

Mean outer-fold and pooled ROC-AUC were reported separately because they summarise performance differently. Mean outer-fold ROC-AUC gives equal weight to every fold, whereas pooled ROC-AUC gives equal weight to every ROI. The two values can therefore differ when the outer folds contain different numbers or distributions of observations.

Precision–recall performance was evaluated using average precision, reported as PR-AUC. This measure describes the relationship between sensitivity and positive predictive value across classification thresholds. Mean outer-fold PR-AUC and pooled out-of-fold PR-AUC were both calculated.

2.11.3 Threshold-Dependent Classification Measures

The evaluated classifiers produced an estimated probability for the malignant class. An ROI was classified as malignant when

$$\hat{p}_i \geq 0.5, \quad (41)$$

and as benign otherwise. The threshold of 0.5 was fixed before final performance evaluation and was not selected using the outer-test predictions.

The resulting predictions were summarised using a confusion matrix. The four confusion-matrix outcomes were:

- true positive (TP): a malignant ROI classified as malignant;

- true negative (TN): a benign ROI classified as benign;
- false positive (FP): a benign ROI classified as malignant;
- false negative (FN): a malignant ROI classified as benign.

Accuracy was calculated as the proportion of all correctly classified ROIs:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}. \quad (42)$$

Sensitivity, also referred to as recall, represented the proportion of malignant ROIs that were correctly identified:

$$\text{Sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \quad (43)$$

Specificity represented the proportion of benign ROIs that were correctly identified:

$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}}. \quad (44)$$

Precision, equivalent to positive predictive value, represented the proportion of predicted malignant ROIs that were malignant:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}. \quad (45)$$

Balanced accuracy was calculated as the mean of sensitivity and specificity:

$$\text{Balanced Accuracy} = \frac{\text{Sensitivity} + \text{Specificity}}{2}. \quad (46)$$

Balanced accuracy was included because ordinary accuracy can be influenced by unequal class sizes.

The F1-score was calculated as the harmonic mean of precision and sensitivity:

$$F_1 = 2 \frac{\text{Precision} \times \text{Sensitivity}}{\text{Precision} + \text{Sensitivity}}. \quad (47)$$

Cohen’s kappa was also calculated to describe agreement between the predicted and observed classes after accounting for the agreement expected by chance.

2.11.4 Fold-Level and Pooled Metric Summaries

ROC-AUC, PR-AUC, accuracy, balanced accuracy, precision, sensitivity, specificity, F1-score, and Cohen’s kappa were first calculated separately for each outer test fold. Their mean and standard deviation across the five outer folds were then reported.

The same measures were also calculated from the combined out-of-fold predictions. Fold-level and pooled values were retained separately because averaging fold-specific metrics is not always equivalent to calculating the same metric after combining all observations.

For metrics that became undefined in a particular fold, such as Cohen’s kappa when only one predicted class was present, the undefined value was recorded as missing and excluded from the corresponding mean calculation.

2.11.5 Probability Calibration

Discrimination measures whether malignant ROIs generally receive higher scores than benign ROIs. Calibration evaluates a different property: whether the predicted probabilities agree with the observed event frequencies.

Probability calibration was assessed using pooled out-of-fold predictions. Predicted probabilities were divided into approximately equal-frequency groups, and the mean predicted probability in each group was compared with the observed proportion of malignant ROIs.

A well-calibrated model produces values close to the diagonal line of perfect agreement. For example, among observations assigned a predicted probability close to 0.70, approximately 70% would be expected to belong to the malignant class.

Calibration error was additionally summarised using the Brier score:

$$\text{Brier} = \frac{1}{n} \sum_{i=1}^n (\hat{p}_i - y_i)^2. \quad (48)$$

The Brier score is the mean squared difference between the predicted probability and the observed

binary outcome. Lower values indicate better probabilistic prediction.

2.11.6 Patient-Clustered Confidence Intervals

Confidence intervals for the pooled out-of-fold metrics were estimated using patient-cluster bootstrap resampling. This procedure accounted for the fact that some patients contributed more than one ROI.

In each bootstrap iteration, patients were sampled with replacement from the complete set of unique patients. When a patient was selected, all ROIs belonging to that patient were included together. Therefore, the dependency among observations from the same patient was preserved.

A total of

$$B = 2000 \quad (49)$$

bootstrap samples were generated. Samples containing only one target class were excluded because ROC-AUC and PR-AUC could not be calculated.

For each metric, the 95% confidence interval was obtained using the percentile method:

$$CI_{95\%} = [Q_{0.025}, Q_{0.975}], \quad (50)$$

where $Q_{0.025}$ and $Q_{0.975}$ are the 2.5th and 97.5th percentiles of the valid bootstrap estimates.

Patient-clustered confidence intervals were calculated for pooled ROC-AUC, PR-AUC, accuracy, balanced accuracy, precision, sensitivity, specificity, F1-score, Cohen’s kappa, and the Brier score.

2.11.7 Model Ranking and Detailed Evaluation

Within each dataset configuration, the candidate pipelines were ranked primarily according to mean outer-fold ROC-AUC. The standard deviation of outer-fold ROC-AUC and the pooled out-of-fold metrics were retained as complementary measures.

For the highest-ranked configurations, balanced accuracy, sensitivity, specificity, precision, F1-

score, Cohen’s kappa, Brier score, and their confidence intervals were also reviewed. These additional measures were used to describe the operating characteristics of the models but did not replace mean outer-fold ROC-AUC as the predefined primary comparison measure.

Detailed visual evaluation of the selected overall model was performed using pooled out-of-fold ROC and precision–recall curves, a confusion matrix, a probability-calibration curve, and fold-level performance summaries.

2.12 Feature-Selection Stability and Model Interpretation

The feature-selection results were examined to determine whether the same predictors were repeatedly selected from different groups of training patients. The selected feature subset was recorded separately for every outer cross-validation fold.

Model interpretation was subsequently performed using Shapley Additive Explanations (SHAP). Feature-selection stability and SHAP importance were treated as two separate properties. Selection stability described the recurrence of a feature across outer folds, whereas SHAP described the contribution of a feature to the output of the fitted interpretation model.

2.12.1 Feature-Selection Stability

Let $\mathcal{F}^{(q)}$ denote the feature subset selected in outer fold q . For feature j , the selection frequency was calculated as

$$f_j = \frac{1}{K_{\text{outer}}} \sum_{q=1}^{K_{\text{outer}}} \mathbb{I}(j \in \mathcal{F}^{(q)}), \quad (51)$$

where $K_{\text{outer}} = 5$, and $\mathbb{I}(\cdot)$ is an indicator function equal to one when feature j was selected in fold q and zero otherwise.

A selection frequency of 1 indicated that the feature was selected in all five outer folds. Frequencies of 0.8, 0.6, 0.4, and 0.2 represented selection in four, three, two, and one outer fold, respectively.

Features were ranked primarily by their selection frequency. This analysis described how consistently the feature-selection pipeline identified the same predictors when trained using different patient subsets. It did not measure the magnitude or direction of the feature’s effect on the classifier output.

The total selection count was also summarised according to radiomic feature family. For multi-source datasets, the source prefix was retained so that recurrence could be assessed separately for features originating from ADC, Pre, Post1, Post2, and T2.

2.12.2 Selection of Recurrent Features for Interpretation

SHAP analysis was restricted to features that recurred across the outer feature-selection folds. A feature was considered recurrent for the interpretation model when it had been selected in at least 40% of the outer folds, with a minimum requirement of two folds:

$$c_j \geq \max(2, \lceil 0.40K_{\text{outer}} \rceil), \quad (52)$$

where c_j is the number of outer folds in which feature j was selected.

Because the analysis used five outer folds, this criterion retained features selected in at least two folds. The recurrent features were ordered according to their selection counts.

If no feature satisfied the recurrence criterion, the implementation used the most frequently selected features as a conservative fallback, rather than including all available numerical predictors.

2.12.3 Selection and Refit of the Interpretation Model

Within the selected experimental run, the classifier with the highest mean outer-fold ROC-AUC was chosen for detailed interpretation. The performance values associated with this model were taken from its nested-cross-validation out-of-fold predictions.

A separate model was then fitted using the complete cohort for SHAP interpretation. Only the

Table 4: Performance measures used to evaluate the machine-learning classifiers. Threshold-dependent measures were calculated using a malignant-class probability threshold of 0.5.

Measure	Preferred value	Interpretation
ROC-AUC	Higher	Ability to rank malignant ROIs above benign ROIs across all classification thresholds.
PR-AUC	Higher	Relationship between sensitivity and positive predictive value, particularly useful when class frequencies are unequal.
Accuracy	Higher	Proportion of all ROIs classified correctly.
Balanced accuracy	Higher	Average of sensitivity and specificity, giving equal importance to both classes.
Sensitivity	Higher	Proportion of malignant ROIs correctly identified.
Specificity	Higher	Proportion of benign ROIs correctly identified.
Precision	Higher	Proportion of predicted malignant ROIs that were malignant.
F1-score	Higher	Combined measure of precision and sensitivity.
Cohen’s kappa	Higher	Agreement between observed and predicted classes beyond chance.
Brier score	Lower	Mean squared error between predicted probabilities and observed outcomes.

recurrent features defined in Section 2.12.2 were used.

Missing values in these features were replaced using their full-cohort medians, and the resulting values were standardised. Classifier hyperparameters selected in the outer folds were aggregated without using outer-test performance. Numerical hyperparameters were combined using their median, whereas categorical hyperparameters were combined using their most frequently selected value.

The resulting classifier was fitted using all available observations. This full-cohort refit was used only to obtain a single interpretable model and was not used to estimate predictive performance. All reported performance measures remained based on the patient-aware out-of-fold predictions described in Section 2.11.

2.12.4 SHAP Analysis

SHAP was used to describe how the recurrent features contributed to the positive-class model output. The positive class corresponded to malignant ROIs.

For models providing `predict_proba`, SHAP

values were calculated for the estimated probability of the malignant class. When probability estimates were unavailable, the continuous decision-function score was used.

For ROI i , the model output was represented as the sum of a baseline value and the contributions of the retained features:

$$h(\mathbf{x}_i) = \phi_0 + \sum_{j=1}^{P_{XAI}} \phi_{ij}, \quad (53)$$

where $h(\mathbf{x}_i)$ is the positive-class model output, ϕ_0 is the baseline output, and ϕ_{ij} is the SHAP contribution of feature j for ROI i .

A positive SHAP value indicated that the feature moved the model output towards the malignant class, whereas a negative SHAP value moved the output towards the benign class. The magnitude of the value described the strength of that contribution for the corresponding ROI.

Global feature importance was calculated as the mean absolute SHAP value:

$$I_j^{\text{SHAP}} = \frac{1}{n} \sum_{i=1}^n |\phi_{ij}|. \quad (54)$$

A larger value of I_j^{SHAP} indicated that feature j

had a larger average influence on the model output. Absolute values were used for global ranking because both positive and negative contributions could be important.

SHAP values were calculated using an independent background masker and the standardised recurrent feature representation. Therefore, the feature values displayed in the SHAP summary plot represented standardised values rather than the original numerical units.

2.12.5 Interpretation Outputs

Two complementary SHAP summaries were generated.

The global importance bar plot ranked features according to their mean absolute SHAP values. This plot described the average magnitude of each feature's contribution but did not show whether higher feature values increased or decreased the malignant-class output.

The SHAP summary dot plot displayed one point for each ROI and feature. The horizontal position represented the SHAP value, while the point colour represented the standardised feature value. This plot therefore showed both the magnitude and direction of the feature contributions across observations.

The feature-selection frequency table, SHAP importance table, summary dot plot, summary bar plot, fitted interpretation model, imputer, scaler, and aggregated hyperparameters were saved to support reproducibility.

SHAP values were interpreted as explanations of the fitted model rather than as evidence of a causal or biological relationship. A high SHAP importance indicated that the classifier relied strongly on a feature; it did not demonstrate that the feature caused the observed class.

2.13 Software Implementation and Reproducibility

The complete computational workflow was implemented in Python using Jupyter notebooks. The notebook metadata recorded Python version 3.13.9 for the final analysis environment.

The main numerical and data-processing operations were implemented using NumPy, pandas, and SciPy. Machine-learning models, cross-validation procedures, preprocessing transformations, and performance measures were implemented using scikit-learn. SMOTE was implemented using imbalanced-learn, and multiple-testing procedures used functions from statsmodels.

Classifier-specific hyperparameter optimisation was performed using Optuna. XGBoost and LightGBM were used for the corresponding gradient-boosted classifiers. SHAP was used for model interpretation, Matplotlib was used to generate figures, and Joblib was used to save fitted model objects and preprocessing components.

A fixed random seed of 100 was used throughout the workflow whenever the corresponding software function supported random-state control. The seed was applied to cross-validation splitting, feature-selection procedures, SMOTE, Optuna sampling, stochastic classifiers, and bootstrap calculations. This ensured that repeated executions using the same software environment and input data produced the same experimental partitions and model settings.

2.13.1 Parameterised Notebook Execution

The modelling code was organised into parameterised analysis notebooks and separate batch-runner notebooks. Dataset-specific information, including the input file, dataset label, output directory, execution mode, and SMOTE comparison settings, was supplied as notebook parameters.

Papermill was used to execute the same analysis notebook repeatedly with different dataset configurations. This avoided maintaining separate copies of the complete modelling code for individual datasets and reduced the risk of introducing implementation differences among experiments.

Separate batch runners were used for the five experimental groups: single-source, pairwise, triple-source, quadruple-source, and complete five-source analysis. Within each group, the runner identified the required datasets, validated their structure, passed the corresponding parameters to

the modelling notebook, and collected the resulting summary files.

Before execution, each batch runner verified that the modelling notebook contained the required parameter cell and batch-safety controls. Dataset files were also checked before model fitting. A run was not started when its input file was missing, its dataset structure was invalid, or the required number of source datasets was not available.

2.13.2 Batch-Safe Execution

The workflow distinguished between interactive and batch execution. During an interactive run, descriptive and publication-oriented plots could be generated for detailed examination of one dataset. During batch execution, report-only plotting cells were disabled automatically.

Disabling these plots did not change preprocessing, feature selection, cross-validation, hyperparameter optimisation, classifier fitting, or performance calculation. It only avoided repeatedly generating computationally expensive figures while processing all dataset configurations.

To avoid instability caused by nested parallel processing, numerical thread counts were restricted during batch execution. Model fitting used one computational worker, and environment variables controlling OpenMP, OpenBLAS, MKL, VecLib, NumExpr, and Joblib parallelism were limited to a single thread.

This serial execution strategy increased runtime compared with unrestricted parallel processing but provided a consistent and stable execution environment for the full experimental grid.

2.13.3 Output Management and Traceability

Each experimental run was assigned a separate output directory. The directory structure preserved the dataset configuration, feature-selection strategy, SMOTE condition, classifier, and cross-validation results.

The workflow saved the following outputs when applicable:

- cleaned and validated dataset information;

- data-quality and dataset-construction reports;
- configuration snapshots;
- outer-fold patient assignments;
- selected feature counts and feature names;
- feature-selection stability tables;
- classifier-specific selected hyperparameters;
- fold-level predictions and performance measures;
- pooled out-of-fold predictions;
- confidence-interval summaries;
- SMOTE versus no-SMOTE comparison tables;
- model-comparison and publication-summary tables;
- fitted interpretation models and preprocessing objects;
- generated publication and diagnostic figures.

Tabular outputs were primarily stored as comma-separated value files, whereas configuration and compact run summaries were stored in JSON format. Fitted models and preprocessing objects were serialised using Joblib. Executed notebooks were retained by the batch runners to provide a record of the code and parameters used for each run.

The batch runners additionally generated execution logs containing the dataset name, input and output paths, execution status, completion time, and any detected error. Failed runs were recorded explicitly and were not silently included in the final result tables.

After completion of each experimental group, the batch runner collected the individual outputs into standardised summary tables. These tables were used to compare datasets, feature-selection strategies, SMOTE conditions, and classifiers under a consistent reporting format.

The same core modelling and reporting procedures were applied across all 31 dataset configurations. Dataset-specific paths and source combinations changed among experiments, whereas the patient-aware validation design, candidate feature counts, feature-selection strategies, classifier set, tuning budget, random seed, and primary performance measure remained fixed.

3 Results

3.1 Dataset Characteristics and Quality-Control Results

The final computational analysis included all 31 planned dataset configurations. These consisted of five single-source, ten pairwise, ten triple-source, five quadruple-source, and one complete five-source dataset.

The five single-source datasets represented 48 unique patients. Several patients contributed more than one ROI. The single-source datasets contained between 114 and 117 ROIs and between 144 and 145 numerical predictor features. Their individual dimensions and class distributions are reported in Table 1.

Across the single-source datasets, the number of benign ROIs ranged from 50 to 52, and the number of malignant ROIs ranged from 64 to 65. The class distributions were therefore similar, but not identical, among the five sources.

Multi-source concatenation increased the number of predictor columns without increasing the number of independent patients. Because only ROIs available in every source of a given combination were retained, the final sample count varied slightly among configurations.

Pairwise datasets contained between 113 and 117 ROIs and between 289 and 290 predictor features. Triple-source datasets contained between 113 and 117 ROIs and between 434 and 435 features. Quadruple-source datasets contained between 113 and 116 ROIs and between 579 and 580 features.

The complete five-source dataset contained 113 matched ROIs, including 50 benign and 63 malig-

nant ROIs, and 724 numerical predictor features. Table 5 summarises the dataset dimensions at each feature-integration level.

A detailed quality-control audit was available for the ADC dataset. The final ADC dataset contained 114 ROIs from 48 patients. The number of ROIs contributed by each patient ranged from 1 to 7, with a median of 2 and a mean of 2.38.

The ADC audit identified 12 missing numerical values distributed across six feature columns. No positive or negative infinite values were detected. No constant or near-zero-variance features were identified in the complete ADC dataset.

The audit also found no fully duplicated rows, duplicated ROI identifiers, or duplicated patient-ROI pairs. Four patients contributed both benign and malignant ROIs. These patients were retained, and all ROIs belonging to the same patient remained together during cross-validation.

Potential outlying values were detected in multiple ADC feature columns. No ROI was excluded solely because it contained an outlying value, and the original finite measurements were retained for the machine-learning analysis.

Partially missing feature values did not lead to removal of the corresponding ROIs. These values were subsequently handled using training-fold median imputation within the patient-aware cross-validation procedure.

3.2 Overall Performance Across the 31 Dataset Configurations

All 31 planned source configurations were evaluated using the same patient-aware nested cross-validation framework. For each source configuration, the highest-ranked pipeline was identified according to mean outer-fold ROC-AUC after comparing the three feature-selection strategies, the two SMOTE conditions, and the ten candidate classifiers.

Figure 2 compares the highest-ranked pipeline obtained for each of the 31 source configurations. The error bars represent the standard deviation of ROC-AUC across the five patient-aware outer folds.

Table 5: Dataset dimensions across the five feature-source integration levels. Multi-source datasets retained only ROIs available in every source included in the corresponding configuration. Feature concatenation increased the number of predictor columns but did not create additional observations.

Integration level	Configurations, n	ROI range	Feature range	Benign range	Malignant range
Single source	5	114–117	144–145	50–52	64–65
Pairwise	10	113–117	289–290	50–52	63–65
Triple source	10	113–117	434–435	50–52	63–65
Quadruple source	5	113–116	579–580	50–52	63–64
All five sources	1	113	724	50	63

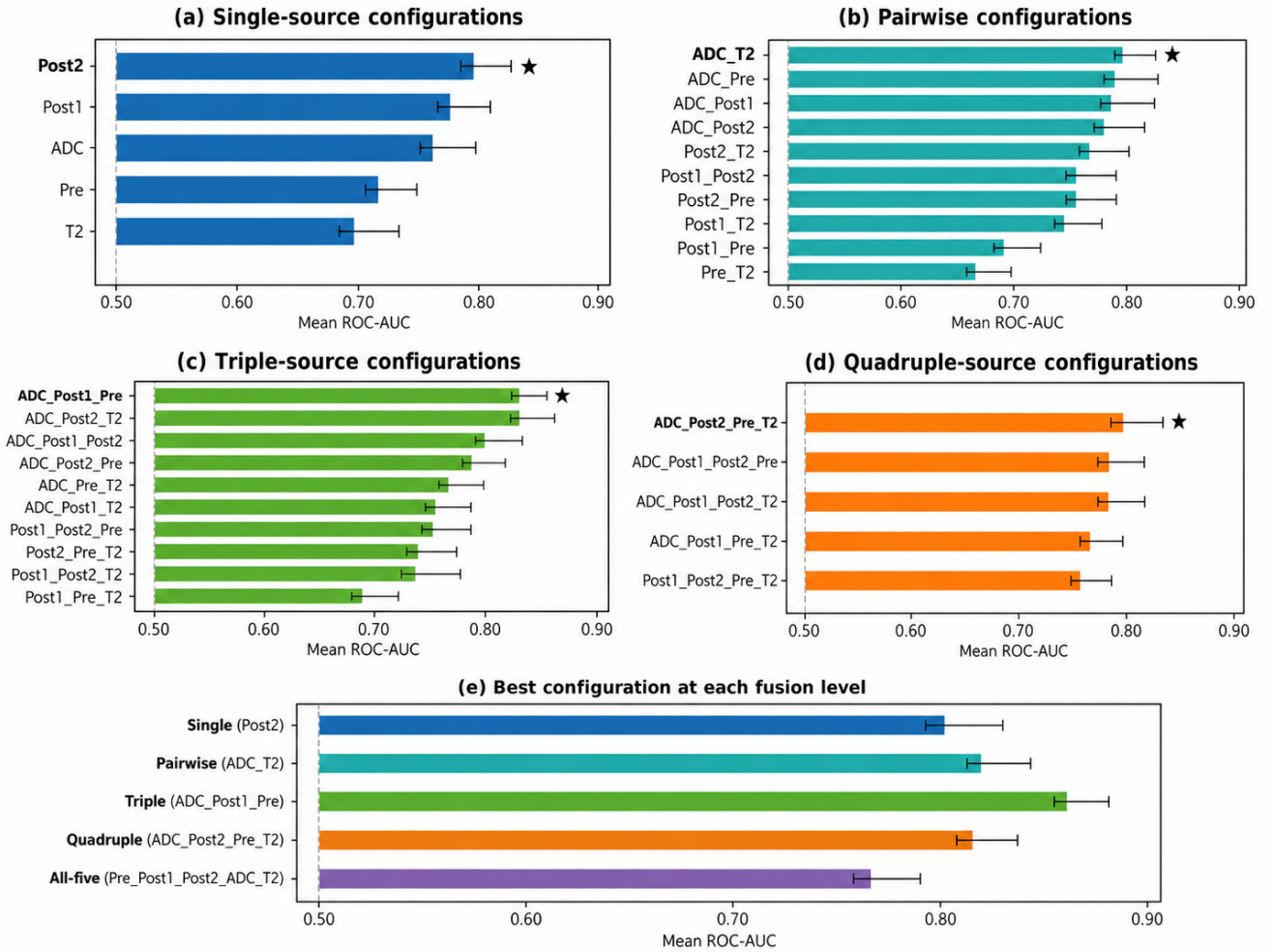


Figure 2: Classification performance across the 31 radiomic source configurations. For each source configuration, the displayed result corresponds to the pipeline with the highest mean ROC-AUC across the five patient-aware outer cross-validation folds. Error bars represent the standard deviation across the five outer folds. (a) Five single-source datasets. (b) Ten pairwise source combinations. (c) Ten triple-source combinations. (d) Five quadruple-source combinations. (e) Highest-ranked configuration at each of the five feature-fusion levels. Configurations within panels (a)–(d) are ordered from the highest to the lowest mean ROC-AUC. The dashed reference line indicates chance-level discrimination at ROC-AUC = 0.50.

As shown in Figure 2, the highest mean ROC-AUC was obtained by the ADC–Post1–Pre triple configuration. The two leading triple configurations achieved mean ROC-AUC values above 0.83. The highest-ranked pairwise and quadruple configurations achieved mean ROC-AUC values of 0.799 and 0.802, respectively, whereas the complete five-source configuration achieved a mean ROC-AUC of 0.747.

The highest-ranked configuration in the complete study was ADC–Post1–Pre. This triple-source dataset was classified using Elastic Net followed by mRMR, Auto- k , no SMOTE, and a linear SVM. It achieved a mean outer-fold ROC-AUC of 0.836 ± 0.088 , together with a pooled out-of-fold ROC-AUC of 0.799. Auto- k selected an average of 9.6 features across the five outer folds.

ADC–Post2–T2 produced a nearly identical mean outer-fold ROC-AUC of 0.834 ± 0.045 . Its fold-level standard deviation was lower than that of ADC–Post1–Pre, while its pooled out-of-fold ROC-AUC was 0.756.

At the other integration levels, Post2 was the highest-ranked single-source dataset, ADC–T2 was the highest-ranked pairwise combination, and ADC–Post2–Pre–T2 was the highest-ranked quadruple combination. The complete five-source configuration had a mean outer-fold ROC-AUC 0.089 lower than that of the highest-ranked triple-source configuration.

Table 6 summarises the highest-ranked pipeline at each feature-fusion level.

3.3 Single-Source Classification Results

The five individual feature sources produced mean outer-fold ROC-AUC values ranging from 0.674 to 0.790. Their relative performance is shown in Figure 2a, and the complete single-source results are summarised in Table 7.

Post2 produced the highest mean outer-fold ROC-AUC among the five single-source datasets. Its highest-ranked pipeline combined Elastic Net followed by mRMR, Auto- k , no SMOTE, and an RBF SVM. This configuration achieved a mean

ROC-AUC of

$$0.790 \pm 0.127, \quad (55)$$

and a pooled out-of-fold ROC-AUC of 0.744. Its mean and pooled PR-AUC values were 0.876 and 0.815, respectively. Auto- k selected an average of 10.8 features across the five outer folds.

Post1 ranked second according to mean outer-fold ROC-AUC, with a value of 0.767 ± 0.098 . Its best pipeline used Elastic Net followed by mRMR, SMOTE, and K-Nearest Neighbours. Post1 achieved the highest pooled ROC-AUC among the single-source datasets, with a value of 0.765, despite ranking below Post2 according to the predefined primary measure.

ADC ranked third, with a mean outer-fold ROC-AUC of 0.764 ± 0.080 . Its best pipeline used mRMR followed by Elastic Net, SMOTE, and a linear SVM. The corresponding pooled out-of-fold ROC-AUC was 0.691.

The highest-ranked Pre pipeline used mRMR followed by Elastic Net, SMOTE, and Logistic Regression. It achieved a mean outer-fold ROC-AUC of 0.698 ± 0.070 and a pooled ROC-AUC of 0.679.

T2 produced the lowest single-source mean outer-fold ROC-AUC, with a value of 0.674 ± 0.049 . Its highest-ranked pipeline used Elastic Net followed by mRMR, SMOTE, and an RBF SVM. Its pooled out-of-fold ROC-AUC was 0.642.

Across the five single-source datasets, the average number of selected features ranged from 7.2 for Post1 to 12.4 for T2. Four of the five highest-ranked single-source pipelines used SMOTE; Post2 was the only single-source winner obtained without SMOTE.

3.4 Pairwise Classification Results

The ten pairwise source combinations produced mean outer-fold ROC-AUC values ranging from 0.665 to 0.799. Their relative performance is shown in Figure 2(b), and the corresponding highest-ranked pipelines are summarised in Table 8.

Table 6: Highest-ranked configuration at each feature-fusion level. Configurations were selected according to mean ROC-AUC across the five patient-aware outer folds. Pooled ROC-AUC was calculated from the combined out-of-fold predictions. EN, Elastic Net; mRMR, minimum redundancy maximum relevance; SVM, support vector machine; SD, standard deviation.

Fusion level	Source configuration	Feature selection	SMOTE	Classifier	Mean AUC $\pm$ SD	Pooled AUC	Mean k
Single	Post2	EN $\rightarrow$ mRMR	No	RBF SVM	0.790 ± 0.127	0.744	10.8
Pairwise	ADC-T2	EN $\rightarrow$ mRMR	No	RBF SVM	0.799 ± 0.084	0.722	10.4
Triple	ADC-Post1-Pre	EN $\rightarrow$ mRMR	No	Linear SVM	0.836 ± 0.088	0.799	9.6
Quadruple	ADC-Post2-Pre-T2	EN $\rightarrow$ mRMR	Yes	RBF SVM	0.802 ± 0.096	0.723	13.2
Five-source	ADC-Pre-Post1-Post2-T2	EN $\rightarrow$ mRMR	No	RBF SVM	0.747 ± 0.117	0.724	12.4

Table 7: Highest-ranked machine-learning pipeline for each single-source dataset. Pipelines were selected according to mean ROC-AUC across the five patient-aware outer folds. PR-AUC was calculated using average precision. EN, Elastic Net; mRMR, minimum redundancy maximum relevance; SVM, support vector machine; KNN, K-Nearest Neighbours; Bal. Acc., balanced accuracy; SD, standard deviation.

Dataset	Feature selection	SMOTE	Classifier	Mean AUC $\pm$ SD	Pooled AUC	Mean PR-AUC	Pooled PR-AUC	Pooled Bal. Acc.	Mean k
Post2	EN $\rightarrow$ mRMR	No	RBF SVM	0.790 ± 0.127	0.744	0.876	0.815	0.669	10.8
Post1	EN $\rightarrow$ mRMR	Yes	KNN	0.767 ± 0.098	0.765	0.798	0.815	0.705	7.2
ADC	mRMR $\rightarrow$ EN	Yes	Linear SVM	0.764 ± 0.080	0.691	0.799	0.720	0.683	8.8
Pre	mRMR $\rightarrow$ EN	Yes	Logistic Regression	0.698 ± 0.070	0.679	0.753	0.740	0.594	10.4
T2	EN $\rightarrow$ mRMR	Yes	RBF SVM	0.674 ± 0.049	0.642	0.708	0.679	0.590	12.4

ADC-T2 achieved the highest mean outer-fold ROC-AUC among the pairwise configurations. Its highest-ranked pipeline used Elastic Net followed by mRMR, Auto- k , no SMOTE, and an RBF SVM. This configuration achieved a mean ROC-AUC of 0.799 ± 0.084 and a pooled out-of-fold ROC-AUC of 0.722. The mean and pooled PR-AUC values were 0.837 and 0.769, respectively. Auto- k selected an average of 10.4 features across the five outer folds.

ADC-Pre ranked second according to mean outer-fold ROC-AUC. Its best pipeline combined Elastic Net followed by mRMR, no SMOTE, and

Logistic Regression. It achieved a mean ROC-AUC of 0.792 ± 0.035 and a pooled ROC-AUC of 0.722. This configuration had the lowest fold-level ROC-AUC standard deviation among the ten pairwise winners.

ADC-Post1 ranked third, with a mean outer-fold ROC-AUC of 0.788 ± 0.050 . Its best pipeline used Elastic Net followed by mRMR, SMOTE, and an RBF SVM. Although it ranked third according to the predefined primary measure, it produced the highest pooled ROC-AUC among the pairwise configurations, with a value of 0.775.

The highest-ranked ADC-Post2 pipeline used

mRMR followed by Elastic Net, SMOTE, and a linear SVM. It achieved a mean outer-fold ROC-AUC of 0.777 ± 0.101 and a pooled ROC-AUC of 0.687.

Post2-T2 achieved a mean outer-fold ROC-AUC of 0.771 ± 0.074 . Its best pipeline used mRMR followed by Elastic Net, SMOTE, and XGBoost, with an average of 8.4 selected features.

The remaining pairwise configurations achieved mean outer-fold ROC-AUC values of 0.763 ± 0.124 for Post1-Post2, 0.756 ± 0.147 for Post2-Pre, 0.742 ± 0.112 for Post1-T2, 0.694 ± 0.138 for Post1-Pre, and 0.665 ± 0.048 for Pre-T2.

The four highest-ranked pairwise configurations all included ADC. Elastic Net followed by mRMR was used by seven of the ten highest-ranked pairwise pipelines, whereas mRMR followed by Elastic Net was used by the remaining three. None of the ten pairwise winners used correlation filtering followed by mRMR.

Seven pairwise winners used SMOTE, whereas three were obtained without SMOTE. RBF SVM was the most frequent winning classifier, appearing in four configurations. The average Auto- k -selected feature count ranged from 5.6 for Post1-Post2 to 11.2 for Post1-T2.

3.5 Triple-Source Classification Results

The ten triple-source configurations produced mean outer-fold ROC-AUC values ranging from 0.682 to 0.836. Their relative performance is shown in Figure 2(c), and the corresponding highest-ranked pipelines are summarised in Table 9.

ADC-Post1-Pre achieved the highest mean outer-fold ROC-AUC among the triple-source configurations and across the complete set of 31 source configurations. Its highest-ranked pipeline used Elastic Net followed by mRMR, Auto- k , no SMOTE, and a linear SVM. It achieved a mean outer-fold ROC-AUC of 0.836 ± 0.088 and a pooled out-of-fold ROC-AUC of 0.799.

The mean and pooled PR-AUC values for this

configuration were 0.879 and 0.832, respectively. Its pooled balanced accuracy was 0.713, and Auto- k selected an average of 9.6 features across the five outer folds.

ADC-Post2-T2 ranked second, with a mean outer-fold ROC-AUC of 0.834 ± 0.045 . Its highest-ranked pipeline also used Elastic Net followed by mRMR, no SMOTE, and a linear SVM. The corresponding pooled ROC-AUC was 0.756. This configuration had a lower fold-level ROC-AUC standard deviation than ADC-Post1-Pre.

ADC-Post1-Post2 ranked third according to mean outer-fold ROC-AUC. Its best pipeline combined Elastic Net followed by mRMR, SMOTE, and Logistic Regression. It achieved a mean ROC-AUC of 0.797 ± 0.075 and a pooled ROC-AUC of 0.790.

The highest-ranked ADC-Post2-Pre pipeline used mRMR followed by Elastic Net, SMOTE, and Logistic Regression. It achieved a mean outer-fold ROC-AUC of 0.793 ± 0.050 and a pooled ROC-AUC of 0.722. Auto- k selected an average of 14.4 features, which was the largest mean feature count among the triple-source winners.

The remaining configurations achieved mean outer-fold ROC-AUC values of 0.767 ± 0.115 for ADC-Pre-T2, 0.754 ± 0.158 for ADC-Post1-T2, 0.753 ± 0.126 for Post1-Post2-Pre, 0.735 ± 0.176 for Post2-Pre-T2, 0.735 ± 0.057 for Post1-Post2-T2, and 0.682 ± 0.074 for Post1-Pre-T2.

The six highest-ranked triple-source configurations all included ADC. Elastic Net followed by mRMR was used by six of the ten highest-ranked pipelines, whereas mRMR followed by Elastic Net was used by the remaining four. None of the triple-source winners used correlation filtering followed by mRMR.

Five triple-source winners used SMOTE, and five were obtained without SMOTE. Linear SVM and Logistic Regression were each the winning classifier for three configurations. Gaussian Naive Bayes was selected for two configurations, whereas RBF SVM and LightGBM were each selected once. The average Auto- k -selected feature count ranged from 5.6 to 14.4.

Table 8: Highest-ranked machine-learning pipeline for each pairwise source configuration. Pipelines were ranked according to mean ROC-AUC across the five patient-aware outer folds. EN, Elastic Net; mRMR, minimum redundancy maximum relevance; SVM, support vector machine; KNN, K-Nearest Neighbours; NB, Gaussian Naive Bayes; SD, standard deviation.

Configuration	Feature selection	SMOTE	Classifier	Mean AUC $\pm$ SD	Pooled AUC	Mean k
ADC-T2	EN $\rightarrow$ mRMR	No	RBF SVM	0.799 $\pm$ 0.084	0.722	10.4
ADC-Pre	EN $\rightarrow$ mRMR	No	Logistic Regression	0.792 $\pm$ 0.035	0.722	9.6
ADC-Post1	EN $\rightarrow$ mRMR	Yes	RBF SVM	0.788 $\pm$ 0.050	0.775	8.0
ADC-Post2	mRMR $\rightarrow$ EN	Yes	Linear SVM	0.777 $\pm$ 0.101	0.687	8.4
Post2-T2	mRMR $\rightarrow$ EN	Yes	XGBoost	0.771 $\pm$ 0.074	0.743	8.4
Post1-Post2	mRMR $\rightarrow$ EN	Yes	RBF SVM	0.763 $\pm$ 0.124	0.744	5.6
Post2-Pre	EN $\rightarrow$ mRMR	Yes	RBF SVM	0.756 $\pm$ 0.147	0.748	10.4
Post1-T2	EN $\rightarrow$ mRMR	Yes	KNN	0.742 $\pm$ 0.112	0.718	11.2
Post1-Pre	EN $\rightarrow$ mRMR	Yes	Logistic Regression	0.694 $\pm$ 0.138	0.687	10.4
Pre-T2	EN $\rightarrow$ mRMR	No	Gaussian NB	0.665 $\pm$ 0.048	0.633	9.6

Table 9: Highest-ranked machine-learning pipeline for each triple-source configuration. Pipelines were ranked according to mean ROC-AUC across the five patient-aware outer folds. EN, Elastic Net; mRMR, minimum redundancy maximum relevance; SVM, support vector machine; NB, Gaussian Naive Bayes; SD, standard deviation.

Configuration	Feature selection	SMOTE	Classifier	Mean AUC $\pm$ SD	Pooled AUC	Mean k
ADC-Post1-Pre	EN $\rightarrow$ mRMR	No	Linear SVM	0.836 $\pm$ 0.088	0.799	9.6
ADC-Post2-T2	EN $\rightarrow$ mRMR	No	Linear SVM	0.834 $\pm$ 0.045	0.756	10.4
ADC-Post1-Post2	EN $\rightarrow$ mRMR	Yes	Logistic Regression	0.797 $\pm$ 0.075	0.790	7.6
ADC-Post2-Pre	mRMR $\rightarrow$ EN	Yes	Logistic Regression	0.793 $\pm$ 0.050	0.722	14.4
ADC-Pre-T2	EN $\rightarrow$ mRMR	Yes	LightGBM	0.767 $\pm$ 0.115	0.720	13.2
ADC-Post1-T2	EN $\rightarrow$ mRMR	Yes	Logistic Regression	0.754 $\pm$ 0.158	0.765	5.6
Post1-Post2-Pre	mRMR $\rightarrow$ EN	Yes	Linear SVM	0.753 $\pm$ 0.126	0.715	11.6
Post2-Pre-T2	EN $\rightarrow$ mRMR	No	RBF SVM	0.735 $\pm$ 0.176	0.707	13.2
Post1-Post2-T2	mRMR $\rightarrow$ EN	No	Gaussian NB	0.735 $\pm$ 0.057	0.706	10.0
Post1-Pre-T2	mRMR $\rightarrow$ EN	No	Gaussian NB	0.682 $\pm$ 0.074	0.695	11.6

3.6 Quadruple-Source Classification Results

The five quadruple-source configurations produced mean outer-fold ROC-AUC values ranging from 0.756 to 0.802. Their relative performance is shown in Figure 2(d), and their highest-ranked pipelines are summarised in Table 10.

ADC–Pre–Post2–T2 achieved the highest mean outer-fold ROC-AUC among the quadruple-source configurations. Its highest-ranked pipeline used Elastic Net followed by mRMR, Auto- k , SMOTE, and an RBF SVM. It achieved a mean outer-fold ROC-AUC of 0.802 ± 0.096 and a pooled out-of-fold ROC-AUC of 0.723.

The mean and pooled PR-AUC values for this configuration were 0.844 and 0.797, respectively. Its pooled balanced accuracy was 0.700, and Auto- k selected an average of 13.2 features across the five outer folds.

ADC–Pre–Post1–Post2 ranked second according to mean outer-fold ROC-AUC. Its highest-ranked pipeline also combined Elastic Net followed by mRMR, SMOTE, and an RBF SVM. It achieved a mean ROC-AUC of 0.790 ± 0.063 and a pooled ROC-AUC of 0.765. Auto- k selected an average of 6.0 features, the smallest mean feature count among the five quadruple-source winners.

ADC–Post1–Post2–T2 ranked third, with a mean outer-fold ROC-AUC of 0.788 ± 0.072 . Its best pipeline used Elastic Net followed by mRMR, SMOTE, and Logistic Regression. This configuration produced the highest pooled ROC-AUC among the quadruple-source datasets, with a value of 0.770. Its pooled PR-AUC was 0.822.

ADC–Pre–Post1–T2 achieved a mean outer-fold ROC-AUC of 0.767 ± 0.063 and a pooled ROC-AUC of 0.725. Its highest-ranked pipeline used Elastic Net followed by mRMR, no SMOTE, and a linear SVM. Auto- k selected an average of 9.6 features.

The Pre–Post1–Post2–T2 configuration produced a mean outer-fold ROC-AUC of 0.756 ± 0.184 and a pooled ROC-AUC of 0.713. Its best pipeline used Elastic Net followed by mRMR, SMOTE, and an RBF SVM. It had the largest fold-level ROC-AUC standard deviation and the largest

average selected feature count, at 15.2 features.

All five highest-ranked quadruple-source pipelines used Elastic Net followed by mRMR. Four of the five winners used SMOTE, whereas the ADC–Pre–Post1–T2 pipeline was obtained without SMOTE.

RBF SVM was the winning classifier for three quadruple-source configurations. Logistic Regression and linear SVM were each selected once. The four configurations containing ADC ranked above the configuration formed from Pre, Post1, Post2, and T2.

3.7 Complete Five-Source Classification Result

The complete five-source configuration combined the ADC, Pre, Post1, Post2, and T2 feature sets. Following ROI matching, this dataset contained 113 observations, including 50 benign and 63 malignant ROIs, and 724 numerical predictor features.

The highest-ranked five-source pipeline used Elastic Net followed by mRMR, Auto- k , no SMOTE, and an RBF SVM. It achieved a mean outer-fold ROC-AUC of 0.747 ± 0.117 and a pooled out-of-fold ROC-AUC of 0.724. The patient-clustered 95% confidence interval for the pooled ROC-AUC was 0.602–0.825.

The mean outer-fold PR-AUC was 0.810, whereas the pooled out-of-fold PR-AUC was 0.792. The corresponding patient-clustered 95% confidence interval for pooled PR-AUC was 0.573–0.907.

The mean outer-fold accuracy was 0.631 ± 0.103 , and the pooled accuracy was 0.619. Mean and pooled balanced accuracy were 0.625 and 0.595, respectively.

At the predefined probability threshold of 0.5, pooled sensitivity was 0.810, pooled precision was 0.622, and the pooled F1-score was 0.703. The pooled Cohen’s kappa was 0.197, and the pooled Brier score was 0.220.

Auto- k selected an average of 12.4 features across the five outer folds. Across these folds, 38 distinct features appeared in at least one selected

Table 10: Highest-ranked machine-learning pipeline for each quadruple-source configuration. Pipelines were ranked according to mean ROC-AUC across the five patient-aware outer folds. EN, Elastic Net; mRMR, minimum redundancy maximum relevance; SVM, support vector machine; SD, standard deviation.

Configuration	Feature selection	SMOTE	Classifier	Mean AUC $\pm$ SD	Pooled AUC	Mean k
ADC-Pre-Post2-T2	EN $\rightarrow$ mRMR	Yes	RBF SVM	0.802 $\pm$ 0.096	0.723	13.2
ADC-Pre-Post1-Post2	EN $\rightarrow$ mRMR	Yes	RBF SVM	0.790 $\pm$ 0.063	0.765	6.0
ADC-Post1-Post2-T2	EN $\rightarrow$ mRMR	Yes	Logistic Regression	0.788 $\pm$ 0.072	0.770	8.0
ADC-Pre-Post1-T2	EN $\rightarrow$ mRMR	No	Linear SVM	0.767 $\pm$ 0.063	0.725	9.6
Pre-Post1-Post2-T2	EN $\rightarrow$ mRMR	Yes	RBF SVM	0.756 $\pm$ 0.184	0.713	15.2

feature subset.

As shown in Figure 2(e), the complete five-source configuration produced the lowest mean outer-fold ROC-AUC among the highest-ranked configurations at the five fusion levels. Its mean ROC-AUC was 0.089 lower than that of the highest-ranked triple-source configuration.

3.8 Comparison of Feature-Selection Strategies

The three sequential feature-selection strategies were compared across all 31 source configurations. For each source configuration, feature-selection strategy, and SMOTE condition, the classifier with the highest mean outer-fold ROC-AUC was first identified. The higher-performing result from the SMOTE and no-SMOTE conditions was then retained. Each feature-selection strategy therefore contributed 31 method-specific mean outer-fold ROC-AUC values.

Figure 3 summarises the distribution of these results and the number of source configurations for which each strategy produced the highest mean outer-fold ROC-AUC.

As shown in Figure 3(a), Elastic Net followed by mRMR produced the highest average and median method-specific mean outer-fold ROC-AUC.

Its mean and median values were 0.754 and 0.760, respectively. The corresponding values were 0.736 and 0.747 for mRMR followed by Elastic Net, and 0.694 and 0.698 for correlation filtering followed by mRMR.

The average mean ROC-AUC obtained with Elastic Net followed by mRMR was approximately 0.018 higher than that obtained with mRMR followed by Elastic Net and 0.060 higher than that obtained with correlation filtering followed by mRMR.

Figure 3(b) shows that Elastic Net followed by mRMR produced the highest method-specific mean ROC-AUC in 20 of the 31 source configurations. mRMR followed by Elastic Net ranked first in the remaining 11 configurations, whereas correlation filtering followed by mRMR did not rank first in any configuration.

At the single-source level, Elastic Net followed by mRMR ranked first for three of the five datasets, whereas mRMR followed by Elastic Net ranked first for two. The corresponding numbers were six and four among the ten pairwise configurations, six and four among the ten triple-source configurations, and four and one among the five quadruple-source configurations. Elastic Net followed by mRMR also produced the highest result for the complete five-source dataset.

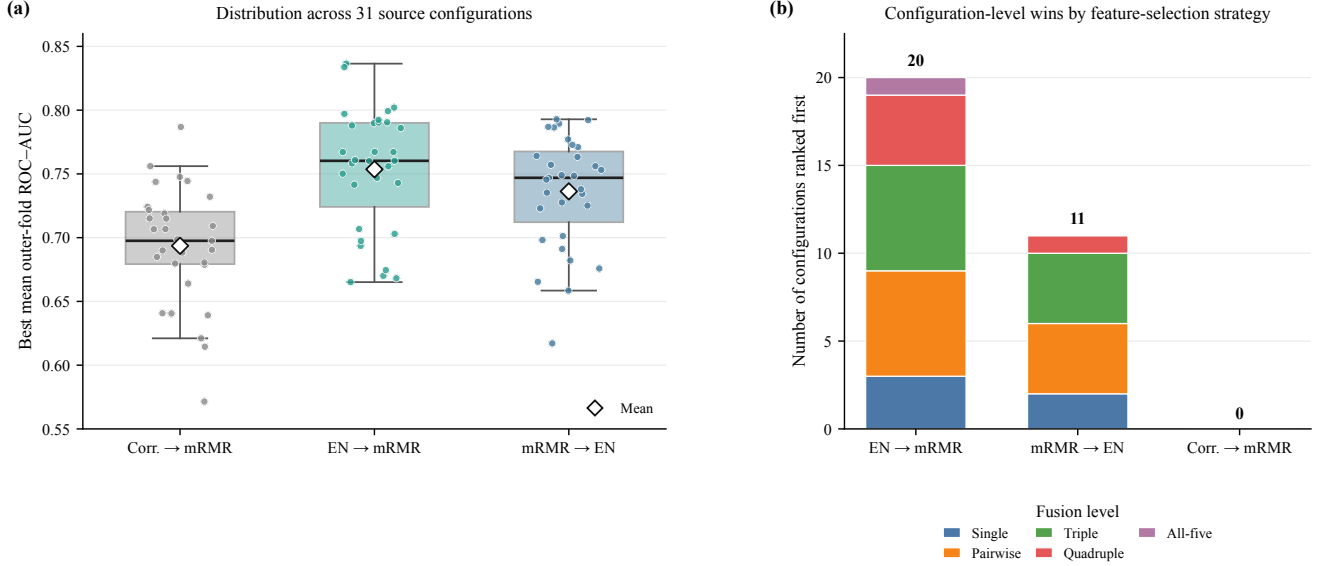


Figure 3: Comparison of the three sequential feature-selection strategies across the 31 source configurations. For each source configuration and feature-selection strategy, the displayed value represents the higher mean outer-fold ROC-AUC obtained after comparison of the SMOTE and no-SMOTE conditions and selection of the highest-performing classifier. (a) Distribution of the 31 method-specific mean outer-fold ROC-AUC values for each feature-selection strategy. Individual points represent source configurations, horizontal lines inside the boxes represent medians, boxes represent interquartile ranges, whiskers represent the remaining non-outlying range, and white diamond markers represent means. (b) Number of source configurations for which each strategy achieved the highest method-specific mean outer-fold ROC-AUC. Stacked bar segments indicate the contributions of single, pairwise, triple-source, quadruple-source, and complete five-source configurations. Numbers above the bars indicate the total number of configuration-level wins. EN, Elastic Net; mRMR, minimum redundancy maximum relevance.

The highest mean outer-fold ROC-AUC obtained using Elastic Net followed by mRMR was 0.836 for ADC–Post1–Pre. The highest values obtained using mRMR followed by Elastic Net and correlation filtering followed by mRMR were 0.793 and 0.787, respectively.

3.9 Effect of SMOTE on Classification Performance

The effect of SMOTE was evaluated separately for all 31 source configurations. For each source configuration and SMOTE condition, the pipeline with the highest mean outer-fold ROC-AUC was identified after comparison of the three feature-selection strategies and ten classifiers. This produced one best no-SMOTE result and one best SMOTE result for each source configuration.

The SMOTE effect for source configuration d was calculated as

$$\Delta \text{AUC}_d = \text{AUC}_{d,\text{SMOTE}} - \text{AUC}_{d,\text{no-SMOTE}}.$$

Positive values indicated higher mean outer-fold ROC-AUC with SMOTE, whereas negative values indicated higher performance without SMOTE. Figure 4 shows the resulting configuration-level differences and the number of wins obtained by each balancing condition.

As shown in Figure 4(a), most configuration-level differences were located close to zero, although both positive and negative effects were observed. The largest positive difference was +0.025 for Post1–Post2–Pre, whereas the largest negative difference was –0.013 for ADC–Post1–Pre.

Figure 4(b) shows that SMOTE produced the higher mean outer-fold ROC-AUC in 19 of the 31 configurations, compared with 12 configurations for no-SMOTE. The two conditions each ranked higher in five pairwise and five triple-source configurations. SMOTE ranked higher in all five quadruple-source configurations, whereas no-SMOTE ranked higher for the complete five-source configuration.

Across the 31 source configurations, the average of the best no-SMOTE mean ROC-AUC

values was 0.757, compared with 0.760 for the best SMOTE results. The average configuration-level difference was therefore +0.003 in favour of SMOTE. The median difference was also +0.003.

SMOTE produced a higher best mean outer-fold ROC-AUC in 19 of the 31 source configurations, whereas no-SMOTE produced the higher result in 12 configurations. No exact ties were observed at the configuration-level comparison.

The direction of the SMOTE effect differed among the feature-fusion levels. SMOTE ranked higher for four of the five single-source datasets, five of the ten pairwise configurations, five of the ten triple-source configurations, and all five quadruple-source configurations. The no-SMOTE condition ranked higher for the complete five-source configuration.

The average SMOTE-minus-no-SMOTE difference was +0.010 for the single-source datasets, +0.0005 for the pairwise configurations, +0.0009 for the triple-source configurations, and +0.006 for the quadruple-source configurations. The difference for the complete five-source configuration was –0.005.

For 26 of the 31 source configurations, the absolute difference between the two balancing conditions was below 0.01 ROC-AUC. Four configurations showed an improvement of at least 0.01 with SMOTE, whereas one configuration showed a decrease of at least 0.01.

The largest positive difference was observed for Post1–Post2–Pre. Its best mean outer-fold ROC-AUC increased from 0.728 without SMOTE to 0.753 with SMOTE, corresponding to a difference of +0.025. The largest negative difference occurred for ADC–Post1–Pre, for which the best mean ROC-AUC decreased from 0.836 without SMOTE to 0.823 with SMOTE, corresponding to a difference of –0.013.

These results show that the direction and magnitude of the observed SMOTE effect varied among source configurations. SMOTE was therefore retained as an experimental pipeline condition rather than applied uniformly to all datasets.

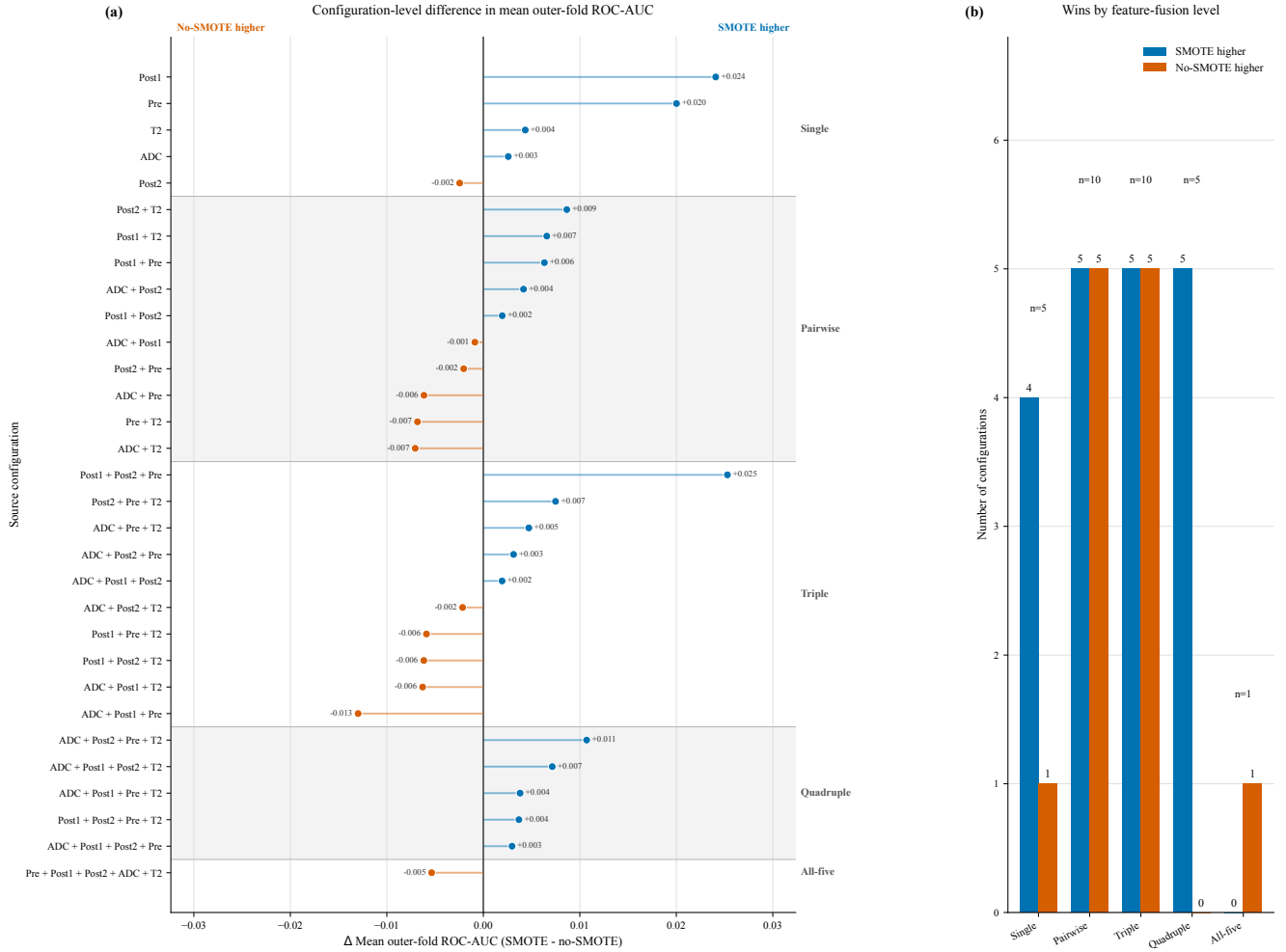


Figure 4: Effect of SMOTE on classification performance across the 31 source configurations. For each source configuration and balancing condition, the displayed result corresponds to the pipeline with the highest mean ROC-AUC after comparison of the three feature-selection strategies and ten classifiers. (a) Configuration-level difference in mean outer-fold ROC-AUC, calculated as the best SMOTE result minus the best no-SMOTE result. Dots represent individual source configurations, and horizontal stems extend from zero to the corresponding difference. Positive values indicate higher performance with SMOTE, whereas negative values indicate higher performance without SMOTE. Configurations are grouped according to feature-fusion level, and alternating background shading separates these groups. (b) Number of source configurations for which SMOTE or no-SMOTE produced the higher mean outer-fold ROC-AUC at each feature-fusion level. Numbers above individual bars indicate configuration-level wins, and annotations above each pair indicate the total number of configurations evaluated at that fusion level.

3.10 Comparison of Machine-Learning Classifiers

The classifiers represented in the highest-ranked pipelines were examined across the 31 source configurations. For each source configuration, one classifier was counted: the classifier belonging to the pipeline with the highest mean outer-fold ROC-AUC after comparison of the three feature-selection strategies and the two SMOTE conditions.

Figure 5 summarises the number of configuration-level selections obtained by each classifier and their distribution across the five feature-fusion levels.

As shown in Figure 5, RBF SVM was the most frequently represented classifier among the highest-ranked pipelines, with 11 configuration-level selections. Logistic Regression and linear SVM followed with seven and six selections, respectively. Together, the two SVM variants accounted for 17 of the 31 highest-ranked pipelines.

Gaussian Naive Bayes was selected in three configurations, and K-Nearest Neighbours was selected in two. XGBoost and LightGBM were each selected once. Random Forest, Extra Trees, and Gradient Boosting did not appear in any configuration-level highest-ranked pipeline.

Seven of the ten candidate classifiers appeared in at least one highest-ranked pipeline. RBF SVM was the most frequently selected classifier, appearing in 11 of the 31 source configurations (35.5%). Logistic Regression ranked second, with seven selections (22.6%), followed by linear SVM with six selections (19.4%).

The two SVM variants together accounted for 17 of the 31 highest-ranked pipelines. Gaussian Naive Bayes was selected in three configurations, and K-Nearest Neighbours was selected in two. XGBoost and LightGBM were each selected once.

Random Forest, Extra Trees, and Gradient Boosting did not appear in the highest-ranked pipeline for any source configuration.

The distribution of classifier selections differed among the feature-fusion levels. At the single-source level, RBF SVM was selected for two datasets, while Logistic Regression, linear SVM,

and K-Nearest Neighbours were each selected once.

Among the ten pairwise configurations, RBF SVM was selected four times, Logistic Regression twice, and linear SVM, Gaussian Naive Bayes, K-Nearest Neighbours, and XGBoost once each.

At the triple-source level, Logistic Regression and linear SVM were each selected for three configurations. Gaussian Naive Bayes was selected twice, while RBF SVM and LightGBM were each selected once.

RBF SVM was selected for three of the five quadruple-source configurations. Logistic Regression and linear SVM were each selected once. RBF SVM was also the highest-ranked classifier for the complete five-source configuration.

These frequencies describe how often each classifier appeared in a configuration-level highest-ranked pipeline. They do not represent the average performance of the classifier across all feature-selection and SMOTE conditions.

3.11 Detailed Performance of the Highest-Ranked Model

The ADC-Post1-Pre configuration produced the highest mean outer-fold ROC-AUC among the 31 evaluated source configurations. Its highest-ranked pipeline combined Elastic Net followed by mRMR, automatic feature-count selection, no SMOTE, and a linear SVM.

Figure 6 presents the pooled out-of-fold discrimination, threshold-dependent classification, and probability-calibration results for this pipeline. All predictions shown in the figure were generated for outer-test observations from patients who were not included in the corresponding model-development partition.

The model achieved a mean outer-fold ROC-AUC of 0.836 ± 0.088 . As shown in Figure 6(a), combining the predictions from the five outer-test folds produced a pooled ROC-AUC of 0.799.

The mean outer-fold PR-AUC was 0.879, and the pooled out-of-fold average precision was 0.832 (Figure 6(b)). The malignant-class prevalence in the pooled out-of-fold dataset was 0.558.

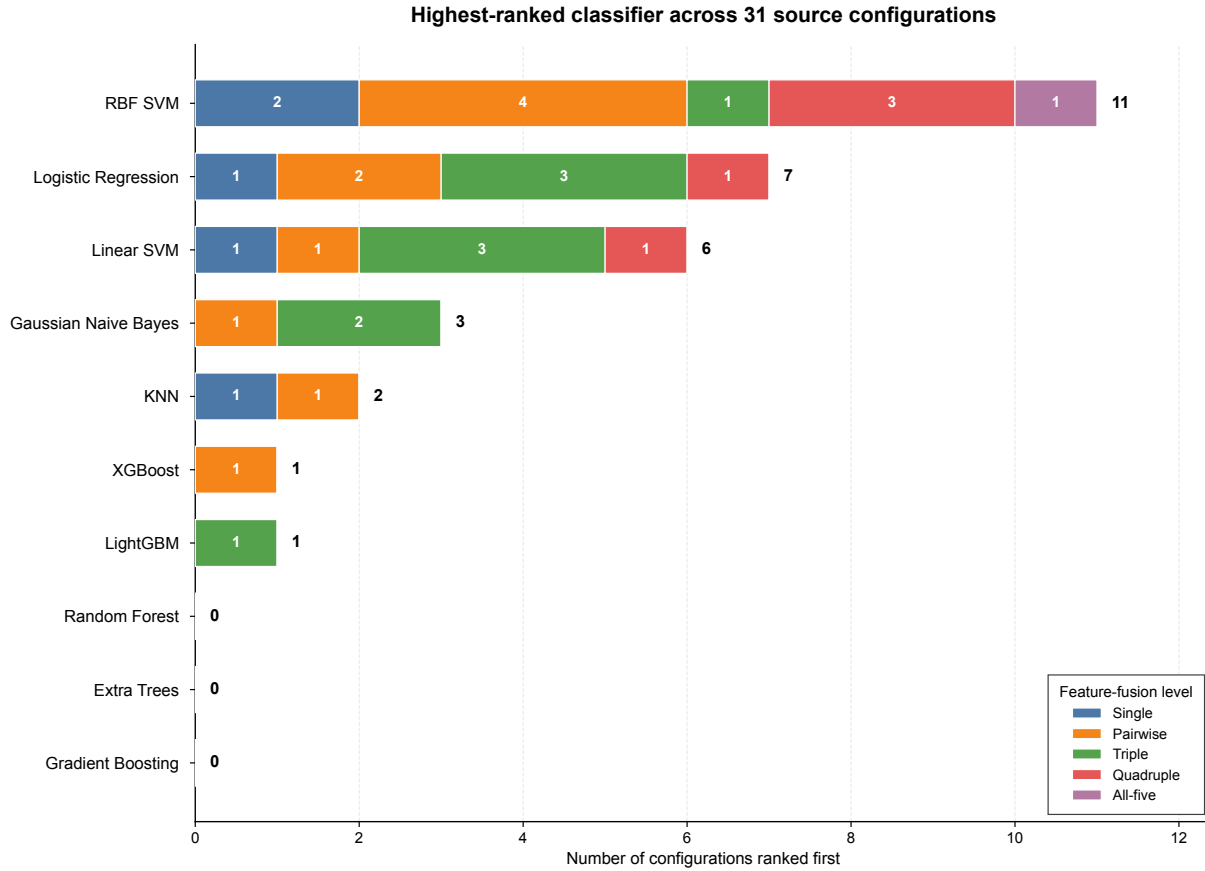


Figure 5: Frequency of classifier selection among the highest-ranked pipelines across the 31 source configurations. For each source configuration, the classifier belonging to the pipeline with the highest mean outer-fold ROC-AUC was counted after comparison of the three feature-selection strategies and the SMOTE and no-SMOTE conditions. Horizontal stacked bars show the numbers of configuration-level selections contributed by single-source, pairwise, triple-source, quadruple-source, and complete five-source analyses. Numbers within coloured segments indicate selections at the corresponding fusion level, whereas numbers at the ends of the bars indicate total selections across all fusion levels. Classifiers with no configuration-level selections are retained to show all ten models evaluated in the computational framework. KNN, K-Nearest Neighbours; SVM, support vector machine.

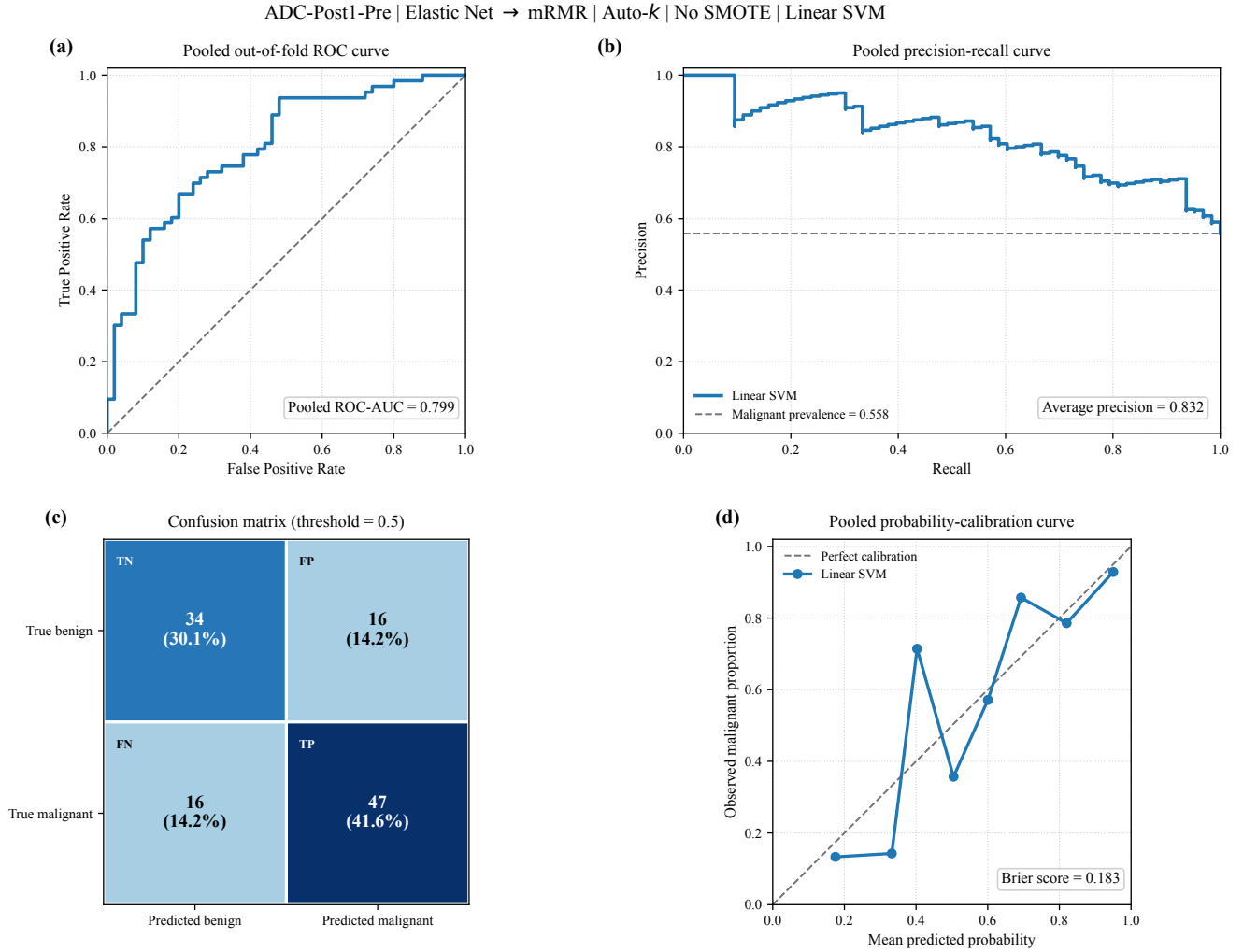


Figure 6: Detailed out-of-fold performance of the highest-ranked machine-learning pipeline. The selected pipeline combined the ADC, Post1, and Pre feature sources, Elastic Net followed by mRMR, automatic feature-count selection, no SMOTE, and a linear SVM. All panels were generated from pooled predictions for the five patient-aware outer-test folds. (a) Pooled receiver operating characteristic curve. The diagonal reference line represents chance-level discrimination. (b) Pooled precision-recall curve. The horizontal reference line represents the malignant-class prevalence of 0.558. (c) Pooled confusion matrix at the predefined malignant-class probability threshold of 0.5. Percentages are calculated relative to all 113 out-of-fold observations. (d) Pooled probability-calibration curve. The diagonal reference line represents perfect calibration. AUC, area under the curve; mRMR, minimum redundancy maximum relevance; SVM, support vector machine.

At the predefined probability threshold of 0.5, the model correctly classified 81 of the 113 ROIs, corresponding to a pooled accuracy of 0.717. The pooled confusion matrix contained 34 true-negative, 16 false-positive, 16 false-negative, and 47 true-positive predictions (Figure 6(c)).

These classification counts corresponded to a sensitivity of 0.746, specificity of 0.680, precision of 0.746, balanced accuracy of 0.713, and F1-score of 0.746. The pooled Cohen’s kappa was 0.426.

The pooled probability-calibration results are shown in Figure 6(d). The corresponding Brier score was 0.183.

Auto- k retained an average of 9.6 features across the five outer folds. The selected feature count ranged from 4 to 18, with a median of 8. Across the five folds, 28 distinct features appeared in at least one selected subset. No feature was selected in all five folds, and the most recurrent feature was selected in four folds. Detailed feature-selection recurrence and model-interpretation results are reported in Section 3.12.

3.12 Feature-Selection Stability and Model Interpretation

Feature-selection stability and SHAP-based model interpretation were examined for the highest-ranked ADC–Post1–Pre pipeline. This pipeline combined Elastic Net followed by mRMR, automatic feature-count selection, no SMOTE, and a linear SVM.

The feature-selection procedure was fitted independently within each of the five patient-aware outer folds. Auto- k retained an average of 9.6 features per outer-fold model, while 28 distinct features appeared in at least one selected subset across the five folds. For the stability analysis, a feature was considered recurrent when it was selected in at least two of the five outer folds.

A total of 13 features satisfied this recurrence criterion (Figure 7(a)). No feature was selected in all five folds. The most recurrent feature was the ADC intensity-histogram median (ADC_IH_IntensityHistogramMedian_Intensity), which was selected in four of the five outer folds.

Five additional features were selected in three folds: the ADC intensity-based median, the Post1 intensity-histogram minimum-gradient grey-level feature, the Post1 intensity-histogram mode, Post1 Compactness 1, and the Post1 normalised zone-size non-uniformity feature. The remaining seven recurrent features were selected in two of the five folds. The recurrent feature set contained predictors derived from ADC and Post1; no Pre-derived feature met the predefined two-fold recurrence criterion.

For model interpretation, a separate linear SVM was refitted using the recurrent feature set and the complete cohort. This refit was performed only to obtain a single model for SHAP analysis and was not used to estimate predictive performance. All reported classification-performance measures remained based on the nested cross-validation out-of-fold predictions.

As shown in Figure 7(b), the ADC intensity-histogram median and ADC intensity-based median had the largest mean absolute SHAP values among the recurrent features. The Post1 intensity-histogram minimum-gradient grey-level feature and Post1 intensity-histogram mode followed them in the global importance ranking. Several Post1 morphological and grey-level size-zone matrix features also contributed to the interpretation model.

The SHAP summary plot (Figure 7(c)) additionally shows the direction and observation-level variability of these contributions. Higher standardised values of the two leading ADC intensity features were generally associated with positive SHAP values, corresponding to a higher fitted malignant-class probability. Higher values of the two leading Post1 intensity-histogram features were also predominantly associated with positive contributions.

In contrast, higher values of several Post1 morphological features, including Compactness and Sphericity, were more frequently associated with negative SHAP contributions. Other recurrent features showed smaller or more heterogeneous contributions across individual ROIs. Feature-selection recurrence and SHAP importance represented different properties of the mod-

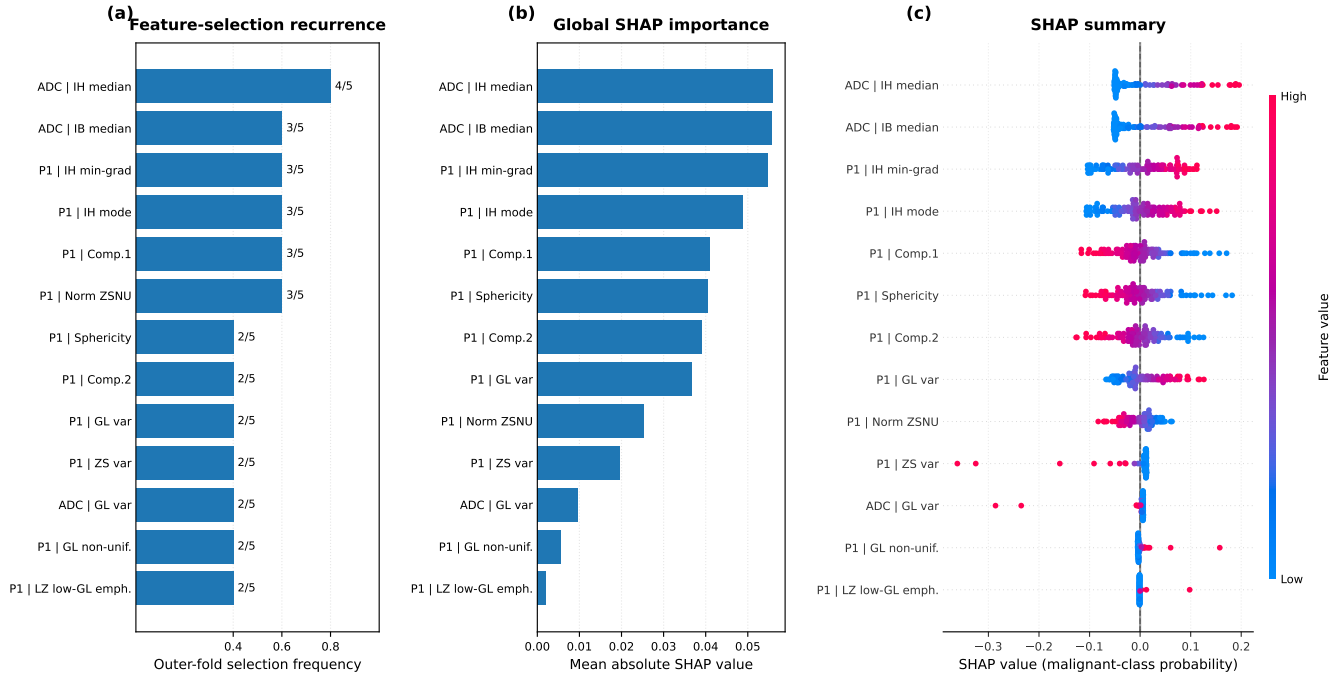


Figure 7: Feature-selection stability and SHAP interpretation of the highest-ranked ADC–Post1–Pre pipeline. The selected pipeline combined Elastic Net followed by mRMR, automatic feature-count selection, no SMOTE, and a linear SVM. (a) Outer-fold recurrence of features selected in at least two of the five patient-aware outer folds. Bar lengths represent the proportion of outer folds in which each feature was retained, and labels indicate the corresponding selection count. (b) Global importance of the recurrent features in the full-cohort interpretation model, quantified using the mean absolute SHAP value. (c) SHAP summary plot showing feature contributions across individual ROIs.

Horizontal position represents the SHAP value for the malignant-class probability, while colour represents the standardised feature value from low to high. Positive SHAP values increase the fitted model output toward the malignant class, whereas negative values decrease it. Feature names were shortened in the figure for readability. P1, Post1; IH, intensity histogram; IB, intensity based; Comp., compactness; GL, grey level; ZS, zone size; ZSNU, zone-size non-uniformity; LZ, large zone; mRMR, minimum redundancy maximum relevance; SHAP, Shapley Additive Explanations; SVM, support vector machine. The full-cohort refit was used for interpretation only; all predictive-performance estimates reported in this study were obtained from patient-aware out-of-fold predictions.

elling pipeline. Selection frequency described how consistently a predictor was retained when the training patients changed, whereas mean absolute SHAP value described its average contribution within the fitted interpretation model. Accordingly, a feature with relatively high recurrence did not necessarily have the highest SHAP importance, and vice versa. SHAP values were interpreted as explanations of the fitted model rather than as evidence of causal or biological relationships.

4 Discussion

4.1 Principal Findings

This study systematically evaluated single- and multi-source radiomic feature representations within a patient-aware machine-learning framework. All 31 non-empty combinations of five MRI-derived feature sources were examined using the same nested cross-validation design, three sequential feature-selection strategies, optional SMOTE, and ten machine-learning classifiers. The main finding was that the highest classification performance was obtained from a selected triple-source representation rather than from complete concatenation of all available feature sources.

The highest-ranked pipeline combined ADC, Post1, and Pre features with Elastic Net followed by mRMR, automatic feature-count selection, no SMOTE, and a linear SVM. It achieved a mean outer-fold ROC-AUC of 0.836 ± 0.088 . A second triple-source configuration, ADC–Post2–T2, achieved a very similar mean ROC-AUC of 0.834 ± 0.045 . In contrast, the complete five-source configuration achieved a lower mean ROC-AUC of 0.747 ± 0.117 . These results indicate that increasing the number of concatenated feature sources did not produce a monotonic improvement in classification performance.

The comparison of feature-selection strategies further showed that the order in which dimensionality-reduction methods were applied was relevant to model performance. Elastic

Net followed by mRMR produced the highest configuration-level result in 20 of the 31 source configurations and also showed the highest average method-specific mean ROC-AUC. mRMR followed by Elastic Net ranked first in the remaining 11 configurations, whereas correlation filtering followed by mRMR did not produce the highest result for any configuration.

The effect of SMOTE was comparatively small and varied among source configurations. Although SMOTE produced the higher configuration-level result more often than the no-SMOTE condition, most differences in mean outer-fold ROC-AUC were close to zero. The highest-ranked model in the complete study was obtained without SMOTE. This finding supports evaluating synthetic oversampling as a model-development choice rather than assuming that it will consistently improve classification performance when class imbalance is limited.

Classifier selection also varied across source configurations. RBF SVM was the most frequently represented classifier among the highest-ranked pipelines, followed by Logistic Regression and linear SVM. Nevertheless, the best overall model used a linear SVM. Thus, the classifier that was selected most frequently across datasets was not necessarily the classifier associated with the single highest ROC-AUC.

The detailed analysis of the highest-ranked ADC–Post1–Pre model showed a pooled out-of-fold ROC-AUC of 0.799 and a pooled PR-AUC of 0.832. At the predefined probability threshold of 0.5, the model produced a balanced accuracy of 0.713 and an F1-score of 0.746. These pooled results were lower than the mean outer-fold ROC-AUC, illustrating that fold-averaged and pooled measures provide related but non-equivalent summaries of predictive performance.

Feature-selection analysis additionally showed that the final models were compact relative to the original feature space but that the identity of the retained features varied across patient groups. The highest-ranked model selected an average of 9.6 features per outer fold, while 28 distinct features appeared across the five folds. Only 13 features were selected in at least two folds, and none

was selected in every fold. This variability indicates that similar predictive performance could be obtained from partially different feature subsets when the model was trained on different groups of patients.

The SHAP analysis of the recurrent feature set showed that the fitted interpretation model relied most strongly on two ADC-derived intensity features, followed by several Post1 intensity, morphological, and grey-level size-zone matrix features. These interpretation results were considered complementary to the stability analysis: selection frequency described how consistently a feature was retained across training subsets, whereas SHAP values described how strongly the refitted model used that feature once included.

Taken together, the findings support three main observations. First, selective multi-source feature integration performed better than complete feature concatenation in this dataset. Second, feature selection and classifier choice interacted with the source representation, such that no single modelling component was uniformly optimal across all configurations. Third, patient-aware validation revealed substantial variation in both predictive performance and selected feature identity across patient subsets, highlighting the importance of evaluating not only peak model performance but also the stability of the complete modelling procedure.

4.2 Effect of Multi-Source Feature Integration

A central objective of this study was to determine whether combining radiomic feature sources improved classification performance relative to analysing the individual sources separately. The results showed that multi-source integration could improve performance, but that the improvement depended strongly on which sources were combined. The relationship between the number of concatenated sources and classification performance was therefore not monotonic.

The best single-source configuration achieved a mean outer-fold ROC-AUC of 0.790, which increased to 0.799 for the highest-ranked pairwise

configuration and to 0.836 for the highest-ranked triple-source configuration. Performance did not continue to increase when additional sources were included. The best quadruple-source configuration achieved a mean ROC-AUC of 0.802, while complete five-source concatenation produced a lower value of 0.747. Thus, the highest performance was obtained from a selected subset of three sources rather than from the complete available feature set.

One possible explanation is the rapid increase in dimensionality created by feature concatenation. The single-source datasets contained approximately 145 candidate predictors, whereas the triple-source datasets contained approximately 435 predictors, the quadruple-source datasets approximately 580, and the complete five-source dataset 724. In contrast, the number of patients remained fixed at approximately 48, and the number of matched ROIs remained close to 113–117. Consequently, adding a feature source increased the number of candidate variables without providing a corresponding increase in the number of independent training patients.

This increasing predictor-to-patient ratio can make model development more difficult. When the candidate feature space becomes larger, the feature-selection procedure must identify a small informative subset from a progressively larger number of alternatives. Additional variables may provide useful complementary information, but they may also introduce redundant or weakly informative predictors. In a limited cohort, distinguishing reliably between these possibilities can become increasingly sensitive to the particular patients included in the training partition.

The Auto- k procedure partially controlled this increase in dimensionality by restricting the final classifier to a relatively small feature subset. For example, the highest-ranked ADC–Post1–Pre model retained an average of only 9.6 features per outer fold despite starting from more than 400 candidate predictors. Nevertheless, feature selection must still operate on the larger initial search space before this compact representation is obtained. The observed reduction in performance with complete feature concatenation therefore can-

not be interpreted simply as a consequence of the final number of variables supplied to the classifier.

The configuration-level results also indicate that the identity of the combined sources was more important than the number of sources alone. Several triple-source combinations performed substantially better than other triple-source combinations, despite having almost identical numbers of observations and predictor variables. Similarly, the ten pairwise configurations covered a broad range of mean ROC-AUC values. This variation suggests that feature-fusion level by itself provides an incomplete description of model performance.

ADC was present in the highest-ranked pairwise, triple-source, and quadruple-source configurations, and the highest-performing triple-source configurations also contained ADC. However, the present analysis does not establish that ADC is independently responsible for the observed improvements. The contribution of one source depends on the other sources with which it is combined, the features retained in each training fold, and the classifier selected for the corresponding pipeline.

The lower performance of the complete five-source model should therefore not be interpreted as evidence that one or more individual sources are inherently uninformative. A source that does not improve performance when concatenated with all other sources may still contribute useful predictors in a smaller combination. Conversely, adding an individually useful source to an already informative feature representation may provide little additional predictive information while increasing the dimensionality of the candidate space.

These findings support evaluating source combinations explicitly rather than assuming that complete feature concatenation is the preferred multi-source strategy. In datasets with a limited number of independent patients, selective feature-source integration can provide a more effective representation than combining every available source. External validation in a larger cohort will be required to determine whether the relative advantage of the selected triple-source configuration is reproducible.

4.3 Influence of Feature-Selection Strategy

Feature-selection strategy had a clear influence on the performance of the evaluated pipelines. Elastic Net followed by mRMR produced the highest configuration-level result in 20 of the 31 source configurations and had the highest average method-specific mean outer-fold ROC-AUC. mRMR followed by Elastic Net ranked first in the remaining 11 configurations, whereas correlation filtering followed by mRMR did not produce the highest-ranked result for any source configuration.

These findings indicate that the order of sequential feature-selection operations was relevant. Elastic Net followed by mRMR and mRMR followed by Elastic Net contained the same two component methods, but they operated on different intermediate feature spaces. The first stage determined which predictors remained available to the second stage and could therefore change the final subset even when the same target feature count was requested.

In the Elastic-Net–mRMR pipeline, Elastic Net first performed a supervised and regularised reduction of the full candidate feature space. The resulting candidate pool was then processed by mRMR, which favoured features with high relevance to the target while penalising redundancy with features that had already been selected. One possible advantage of this ordering is that the second-stage mRMR procedure operated on a smaller candidate space that had already undergone supervised regularised screening.

The reverse pipeline began by applying mRMR directly to the complete feature space and then used Elastic Net to rank the reduced candidate pool. This strategy remained competitive and produced the highest-ranked result in approximately one third of the source configurations. Its performance therefore shows that there was no universally optimal ordering of the two selectors. Nevertheless, the more frequent wins and higher average ROC-AUC observed for Elastic Net followed by mRMR suggest that this ordering was better suited to the present datasets.

Correlation filtering followed by mRMR

showed lower performance than the two Elastic-Net-based strategies. The correlation stage removed features when their absolute pairwise Pearson correlation exceeded 0.90 and retained one representative using direction-independent univariate ROC-AUC. This procedure can efficiently reduce strongly redundant predictors, but it considers pairwise linear correlation before the subsequent multivariable selection stage. In a high-dimensional radiomic feature space, correlated predictors may not be interchangeable with respect to their contribution in combination with other variables. Consequently, an early correlation filter may remove predictors that would otherwise contribute useful information to a multivariable model.

This interpretation should remain cautious. The present experiments did not isolate the individual causal contribution of each feature-selection operation, and the three strategies were compared within the same finite cohort. Their relative ordering may therefore depend on the specific feature distributions, patient composition, candidate feature counts, and cross-validation partitions used in this study.

The results also show that feature-selection performance should not be judged only by the single highest ROC-AUC obtained in the study. Although Elastic Net followed by mRMR produced the overall highest-ranked model, mRMR followed by Elastic Net remained preferable for 11 source configurations. This variation reinforces the need to treat feature selection as part of the complete modelling pipeline rather than as a fixed preprocessing operation that can be selected independently of the input representation.

Overall, the results support the use of sequential supervised feature selection for these high-dimensional datasets, with Elastic Net followed by mRMR providing the most consistently favourable results among the three evaluated strategies. Confirmation in an independent cohort would be required before considering this ordering generally preferable for radiomic classification tasks.

4.4 Effect of SMOTE and Class Imbalance

SMOTE did not produce a uniform improvement in classification performance. At the source-configuration level, the best SMOTE pipeline achieved a higher mean outer-fold ROC-AUC in 19 of the 31 configurations, whereas the best no-SMOTE pipeline ranked higher in 12. However, the magnitude of most differences was small. Therefore, the larger number of SMOTE wins should not be interpreted as evidence of a substantial general improvement in discrimination.

This distinction between win frequency and effect size is important. A source configuration was counted as a SMOTE win even when the difference in mean ROC-AUC was only a few thousandths. Consequently, a larger number of configuration-level wins can coexist with an average ROC-AUC difference close to zero. In the present experiments, the largest positive configuration-level difference was only 0.025, while the largest negative difference was 0.013.

The limited overall effect of SMOTE is also consistent with the class distribution of the datasets. The imbalance between benign and malignant ROIs was relatively modest. For example, the complete five-source dataset contained 50 benign and 63 malignant ROIs. Synthetic oversampling therefore modified a dataset that was not severely imbalanced. Under these conditions, the additional synthetic observations may alter the fitted decision boundary without producing a substantial improvement in the overall ranking of unseen observations.

An important distinction is that ROC-AUC is threshold independent, whereas several other classification measures depend on the selected decision threshold. SMOTE changes the training distribution and can therefore alter classifier coefficients, support vectors, neighbourhood relationships, or tree structures. These changes may affect sensitivity and specificity at a fixed threshold even when the relative ranking of observations, and therefore ROC-AUC, changes only slightly.

The more detailed paired analyses performed in this study were consistent with this interpreta-

tion. Across dataset and feature-selection comparisons, SMOTE had almost no average effect on mean outer-fold ROC-AUC but tended to increase pooled balanced accuracy. This improvement was accompanied by a shift in class-specific performance: specificity generally increased, whereas sensitivity tended to decrease. Because the positive class in this study represented malignant ROIs, this pattern corresponds to improved recognition of the smaller benign class at the cost of some reduction in malignant-class sensitivity.

This trade-off also explains why the effect of SMOTE depended on the performance measure considered. A model may achieve higher balanced accuracy or specificity while showing little change in ROC-AUC, and an improvement in precision may occur together with a reduction in sensitivity. SMOTE should therefore not be judged using only whether it increases one threshold-dependent metric.

The effect of oversampling also varied with feature-selection strategy and source configuration. This is expected because SMOTE generates synthetic observations by interpolation in the selected feature space. The geometry of that space changes when different predictors are retained, and the same synthetic-sampling procedure may therefore modify different classifiers in different ways. For example, synthetic samples can directly change local neighbourhoods for KNN, the position of support vectors for SVMs, and candidate partitions for tree-based models.

Importantly, the two highest-ranked triple-source configurations, ADC-Post1-Pre and ADC-Post2-T2, were both obtained without SMOTE. The complete five-source winner was also a no-SMOTE pipeline. These findings show that synthetic oversampling was not required to achieve the strongest discrimination observed in the study.

Conversely, SMOTE was selected in several successful pipelines, including all five configuration-level winners in the quadruple-source analysis. Its usefulness was therefore configuration-dependent rather than uniformly absent. The results do not support removing SMOTE from consideration; instead, they support

evaluating it within the model-development procedure and retaining it only when it improves the predefined optimisation objective.

SMOTE and decision-threshold adjustment should also be regarded as different operations. SMOTE changes the training data before model fitting and can alter the fitted decision function itself. Threshold adjustment does not refit the model and instead changes only the point at which a continuous model output is converted into a class label. If the primary objective is to change the sensitivity–specificity trade-off rather than improve ranking performance, threshold optimisation may therefore represent a distinct and potentially more direct strategy.

In the present study, mean outer-fold ROC-AUC was predefined as the primary model-ranking measure. According to this criterion, SMOTE provided no consistent or substantial general advantage. Its effects on balanced accuracy and class-specific error rates remain relevant, but these complementary changes should be considered in relation to the intended application rather than interpreted as evidence that synthetic oversampling is universally beneficial.

4.5 Classifier Performance and Model Complexity

Classifier performance varied across source configurations, and no single classifier was optimal for every dataset. Nevertheless, the highest-ranked pipelines were concentrated among a relatively small subset of the ten evaluated models. RBF SVM was the most frequently selected classifier, appearing in 11 of the 31 configuration-level winners, followed by Logistic Regression in seven configurations and linear SVM in six. Together, the two SVM variants accounted for 17 of the 31 highest-ranked pipelines.

The frequent selection of SVM-based models is compatible with the structure of the present classification problem. After feature selection, each outer-fold classifier was fitted using a relatively small number of predictors but a limited number of independent patients. Margin-based clas-

sifiers can provide flexible decision boundaries while controlling model complexity through regularisation. The linear and RBF variants also allowed the analysis to test whether the selected feature representation was more appropriately separated by a linear or nonlinear boundary.

The overall highest-ranked model used a linear SVM rather than the more frequently selected RBF SVM. This finding is important because classifier win frequency and peak predictive performance measure different properties. RBF SVM was selected most often across the 31 source configurations, but the highest individual mean outer-fold ROC-AUC was obtained using a linear decision boundary for the ADC-Post1-Pre representation.

A linear classifier performing well after feature selection may indicate that the retained variables provide a representation in which substantial class separation can be captured without a highly complex decision function. However, this should not be interpreted as evidence that the underlying relationships among all original radiomic features are strictly linear. The feature-selection stages were supervised and could remove variables that were less useful for the final classification problem, thereby changing the geometry of the input space presented to the classifier.

Logistic Regression was also frequently represented among the highest-ranked pipelines. Both Logistic Regression and linear SVM apply regularised linear decision functions, although they optimise different objectives and produce different probability estimates. Their frequent selection suggests that comparatively simple regularised models remained competitive after dimensionality reduction.

In contrast, the tree-based ensemble models were less frequently represented among the configuration-level winners. XGBoost and LightGBM were each selected once, while Random Forest, Extra Trees, and Gradient Boosting did not appear in any final configuration-level winning pipeline. This result does not establish that tree-based methods are generally unsuitable for radiomic classification. Rather, within the present dataset size, selected feature representations, hyperparameter search spaces, and nested

cross-validation procedure, they did not achieve the highest mean outer-fold ROC-AUC for the evaluated source configurations.

One possible explanation is that flexible ensemble models require sufficient training data to estimate complex interactions reliably. In the present study, the number of independent patients was small relative to the original number of candidate predictors. Although feature selection substantially reduced the dimensionality before classifier fitting, the effective training sample in each outer and inner fold remained limited. Under such conditions, additional model flexibility may not necessarily translate into improved performance on previously unseen patients.

The restricted hyperparameter-optimisation budget should also be considered when interpreting classifier comparisons. Each Optuna study evaluated ten candidate hyperparameter configurations. This provided a consistent computational budget across classifiers but was not intended to exhaustively explore every possible region of the search space. Models with larger or more complex hyperparameter spaces, particularly boosted-tree methods, may therefore have been more sensitive to the limited tuning budget than models with only one or two principal hyperparameters.

K-Nearest Neighbours and Gaussian Naive Bayes occasionally produced the highest-ranked result for individual source configurations, showing that comparatively simple non-linear or probabilistic approaches could also be competitive for particular feature representations. Their limited number of wins, however, indicates that their performance was more configuration specific.

The classifier results therefore reinforce the importance of evaluating the classifier together with the preceding feature representation rather than treating model choice as an isolated decision. The optimal classifier depended on the source combination, selected feature subset, and class-balancing condition. In the present study, regularised linear and kernel-based models provided the most frequent high-performing solutions, while the single best-performing pipeline used the simpler linear SVM.

4.6 Feature Stability and Model Interpretation

The feature-stability analysis provided information that was not captured by predictive-performance measures alone. Although the highest-ranked ADC–Post1–Pre pipeline retained an average of only 9.6 features in each outer-fold model, the identity of these features varied when the training patients changed. Across the five outer folds, 28 distinct features were selected at least once, while only 13 were retained in two or more folds and no feature was selected in all five folds.

This finding highlights an important distinction between model performance and feature-selection stability. A pipeline can maintain relatively strong classification performance while selecting partially different subsets of correlated or similarly informative predictors in different training samples. In high-dimensional radiomic datasets, several features may contain overlapping information, allowing different subsets to support similar predictive functions.

The limited recurrence observed in the present study may also reflect the small number of independent patients relative to the original feature space. Each change in the outer-training population alters the data available to the supervised feature-selection procedure. With hundreds of candidate predictors and a comparatively small training cohort, modest changes in the training observations can change feature rankings and therefore the subset passed to the final classifier.

Sequential feature selection may further contribute to this variability. In the highest-ranked pipeline, Elastic Net first reduced the candidate space and mRMR subsequently selected a compact subset from the retained predictors. A feature excluded during the first stage was no longer available to the second stage, and small changes in Elastic-Net coefficients across training folds could therefore propagate to the final selected subset. Similarly, the mRMR redundancy term depends on features that have already been selected and can lead to different choices among correlated candidates.

The stability results therefore suggest that indi-

vidual selected features should be interpreted more cautiously than the predictive performance of the complete pipeline. A feature that appeared in only one outer fold may still have contributed to a valid predictive model in that fold, but its limited recurrence provides weaker evidence that it represents a robust predictor across patient subsets. Conversely, features selected repeatedly across several folds provide stronger evidence of selection stability, although recurrence alone does not measure their contribution to the classifier output.

This distinction was apparent when feature-selection frequency was compared with SHAP importance. The most recurrent ADC intensity-histogram median feature also had the largest mean absolute SHAP value in the interpretation model, but recurrence and SHAP importance were not identical rankings for all features. Selection frequency measures how consistently a variable survives the model-development procedure, whereas SHAP measures its contribution after the variable has been included in a fitted model.

The recurrent feature set for the highest-ranked pipeline contained features derived from ADC and Post1. No Pre-derived feature satisfied the predefined requirement of selection in at least two outer folds. This observation should not be interpreted as showing that the Pre source was uninformative. The highest-performing source configuration itself included Pre, and different Pre-derived features could have contributed in individual folds without recurring often enough to enter the stability-based interpretation set. In addition, the contribution of one source may be expressed through interactions or through changes in which features from the other sources are selected.

The SHAP analysis indicated that the full-cohort interpretation model relied most strongly on two ADC-derived intensity features, followed by several Post1 intensity, morphological, and grey-level size-zone matrix features. Higher values of the leading ADC intensity features were generally associated with positive SHAP contributions in the fitted model, whereas several Post1 morphological features showed the opposite tendency. These directions describe how the fitted classifier used the standardised predictors and should not be in-

terpreted as direct biological or causal effects.

An additional limitation is that SHAP values were calculated from a separate model refitted using the complete cohort and the recurrent feature set. This refit was necessary to provide one common model for interpretation, but its predictions were not used to estimate classification performance. The reported ROC-AUC, PR-AUC, and threshold-dependent metrics remained based on patient-aware out-of-fold predictions. Consequently, the SHAP analysis should be viewed as an explanation of the final fitted representation rather than as an independent validation of feature importance.

Overall, the stability and interpretation analyses support reporting the selected predictors as a recurrent feature set rather than as a single fixed signature. The observed variability indicates that external validation should assess not only whether predictive performance is reproduced, but also whether the same feature sources, feature families, and individual predictors continue to be selected in an independent cohort.

4.7 Strengths and Limitations

Several aspects of the computational design strengthen the reliability of the present analysis. First, model evaluation was patient aware. Although individual ROIs were used as prediction units, all ROIs from the same patient were assigned to the same cross-validation partition. This prevented observations from the same patient from appearing simultaneously in training and test data and reduced the risk of patient-level information leakage.

Second, model development was performed within a nested cross-validation framework. Hyperparameter optimisation, feature selection, feature-count selection, preprocessing, and the optional application of SMOTE were restricted to the training data available within the corresponding cross-validation stage. The outer folds were therefore not used to fit these components before prediction. This design provides a more realistic estimate of performance on unseen patients than procedures in which feature selection or

hyperparameter optimisation is performed before cross-validation.

Third, the study evaluated the complete set of 31 non-empty combinations of the five available radiomic feature sources using a common computational framework. The same three feature-selection strategies, SMOTE comparison, classifier set, optimisation procedure, and primary evaluation criterion were applied across the source configurations. This systematic design allowed differences among single- and multi-source representations to be evaluated under largely consistent modelling conditions rather than by comparing independently developed pipelines.

The analysis also extended beyond a single discrimination metric. Mean outer-fold ROC-AUC was predefined as the primary model-ranking criterion, while pooled out-of-fold ROC-AUC, PR-AUC, threshold-dependent classification metrics, calibration, feature recurrence, and SHAP-based interpretation provided complementary information. In particular, the stability analysis showed that a high-performing model can remain predictive while selecting partially different feature subsets across training samples, which would not have been apparent from ROC-AUC alone.

These strengths should be considered together with several important limitations. The principal limitation is the small number of independent patients. The datasets contained approximately 48 patients, despite including more than 100 ROIs. Because cross-validation was grouped at the patient level, the effective number of independent units available for model development remained limited. This is particularly important for the multi-source datasets, in which several hundred candidate predictors were available before feature selection.

The use of multiple ROIs per patient also means that the reported classification results are ROI-level rather than patient-level performance estimates. Patient grouping prevents leakage across cross-validation folds, but it does not convert multiple observations from one patient into independent patients. Consequently, the number of ROIs should not be interpreted as the effective sample size of the study.

A further limitation is the absence of an independent external test cohort. Nested cross-validation reduces optimisation bias within each model-development procedure, but all source configurations and modelling strategies were ultimately compared using the same study cohort. The highest-ranked pipeline was selected from a large number of evaluated alternatives, and its observed performance may therefore include some degree of selection optimism. External validation is required to determine whether the ranking of source configurations and the performance of the ADC–Post1–Pre pipeline are reproducible in independent data.

The computational search was also necessarily constrained. Ten classifiers, three feature-selection strategies, two SMOTE conditions, multiple candidate feature counts, and 31 source configurations already produced a large model-development space. Optuna was therefore limited to ten trials for each optimisation study. This ensured a consistent and computationally manageable search procedure, but it should not be interpreted as exhaustive hyperparameter optimisation. Classifiers with larger or more complex hyperparameter spaces may have benefited from a larger optimisation budget.

The study used direct feature concatenation to construct multi-source datasets. This approach provides a transparent way to compare all source combinations, but it does not explicitly model relationships between sources. Alternative integration methods, including source-specific dimensionality reduction, late fusion, representation learning, or other multimodal modelling strategies, could produce different results and were outside the scope of the present analysis.

Feature-selection stability represents another limitation. The highest-ranked model retained a compact number of predictors in each outer fold, but the selected feature identities varied across folds, and no individual feature was selected in all five folds. The SHAP results should therefore be interpreted as descriptions of the fitted model rather than evidence of a fixed or externally validated feature signature.

In addition, the SHAP analysis required a separate full-cohort refit using the recurrent feature

set to obtain one common model for interpretation. This model was used only to explain fitted prediction behaviour and not to estimate predictive performance. Nevertheless, its SHAP values describe a model fitted to the complete available cohort and therefore do not constitute independent evidence that the same feature contributions will be reproduced in new patients.

Calibration and threshold-dependent metrics should also be interpreted cautiously. The limited cohort size restricts the number of observations available within calibration bins, and the fixed probability threshold of 0.5 was not optimised for a particular clinical operating point. Sensitivity, specificity, precision, and related measures could therefore change if a different decision threshold were selected in an independent development cohort.

Finally, the present work was designed as a computational evaluation of radiomic machine-learning strategies. It does not establish causal or biological interpretations of individual radiomic features, nor does it demonstrate clinical utility. The reported results should therefore be considered evidence regarding the comparative behaviour of the evaluated modelling pipelines within the available cohort, rather than confirmation of a deployable diagnostic model.

Despite these limitations, the combination of patient-aware nested cross-validation, training-only preprocessing, systematic comparison of source combinations, and out-of-fold performance evaluation provides a controlled framework for identifying candidate modelling strategies for subsequent independent validation.

4.8 Future Work and Generalisability

The next priority is independent validation of the highest-ranked ADC–Post1–Pre pipeline. Although patient-aware nested cross-validation was used throughout the present study, all model development and configuration comparisons were performed within a single cohort. Evaluation in an independent dataset is therefore required to determine whether the observed discrimination and the relative advantage of the selected triple-source

representation are reproducible.

External validation should preserve the patient-level grouping used in the present analysis and should evaluate the complete modelling pipeline rather than only the final classifier. This includes the preprocessing operations, feature-selection sequence, automatic feature-count selection, and classifier specification. Reapplying feature selection within the external development process will also be important for determining whether the recurrent features identified in the present cohort remain stable when the patient population changes.

A larger cohort would additionally allow more precise estimation of predictive performance. In the present study, the limited number of independent patients contributed to variability across outer folds and restricted the precision of calibration and threshold-dependent performance estimates. Increasing the number of patients, rather than only the number of ROIs, would provide a stronger basis for estimating generalisation performance and feature-selection stability.

Future studies should also investigate whether the relative advantage of selective source integration persists in larger datasets. The present results favoured specific triple-source combinations over complete five-source concatenation, but this relationship may change as the number of training patients increases. A larger cohort may allow models to make more effective use of additional predictors and may reduce the instability associated with selecting a small feature subset from a very high-dimensional candidate space.

The present multi-source analysis used direct feature concatenation, which was intentionally chosen for transparency and systematic comparison of all 31 source combinations. Future computational work could compare this strategy with alternative fusion methods. Examples include source-specific feature selection followed by late fusion, separate source-level classifiers with prediction-level combination, or representation-learning approaches designed to preserve source-specific information before integration. Such comparisons would help determine whether the lower performance observed with complete concatenation resulted from the information content of

the sources themselves or from the way in which the feature spaces were combined.

More extensive hyperparameter optimisation may also be useful, particularly for classifiers with relatively large search spaces. The present ten-trial Optuna budget provided a consistent search procedure across a large number of pipelines, but it was not intended to identify the global optimum for every classifier. A future validation study could allocate a larger optimisation budget to a smaller set of candidate pipelines identified from the current screening analysis.

Model calibration and decision-threshold selection represent further areas for investigation. The present study used a fixed probability threshold of 0.5 for threshold-dependent metrics and did not optimise a threshold for a specific operating objective. In future work, threshold selection could be performed exclusively within training data and then evaluated on independent patients. This would allow sensitivity, specificity, precision, and related measures to be adapted to a predefined application requirement without introducing test-set optimisation.

The feature-stability analysis should also be repeated in independent data. Reproducibility of individual feature names would provide the strongest evidence of a stable signature, but stability can also be examined at broader levels, including feature source and feature family. This distinction may be useful when several correlated radiomic variables represent similar information and individual feature identities vary across samples.

Similarly, SHAP-based model interpretation should be regarded as provisional until the model is evaluated externally. Future analyses should determine whether the features with high SHAP importance in the present interpretation model retain similar importance and direction of contribution in independently fitted models. Agreement between predictive performance, feature recurrence, and model interpretation across datasets would provide substantially stronger evidence of model robustness than any of these analyses alone.

Finally, future work should distinguish model-development performance from prospective or real-world performance. A computational pipeline

that performs well under nested cross-validation and external retrospective validation would still require additional evaluation before practical deployment. Such evaluation should include robustness to dataset shift, acquisition differences, missing feature sources, calibration changes, and the definition of an appropriate decision threshold for the intended use.

Overall, the present study should be viewed as a systematic model-development and comparison framework that identifies promising source combinations and modelling strategies for subsequent validation. The ADC–Post1–Pre pipeline provides the strongest candidate from the evaluated search space, but confirmation in larger and independent cohorts is required before its performance, feature-selection pattern, or interpretability results can be considered generalisable.

5 Conclusion

This study systematically compared all 31 non-empty combinations of five MRI-derived radiomic feature sources within a patient-aware nested cross-validation framework. The results showed that increasing the number of concatenated feature sources did not consistently improve classification performance. Instead, selective multi-source integration produced the strongest results, with the highest-ranked pipeline combining ADC, Post1, and Pre features with Elastic Net followed by mRMR, automatic feature-count selection, no SMOTE, and a linear SVM. This pipeline achieved a mean outer-fold ROC-AUC of 0.836 ± 0.088 and a pooled out-of-fold ROC-AUC of 0.799.

Across the broader model comparison, feature-selection strategy, class-balancing choice, and classifier performance depended on the source representation, supporting their evaluation as components of the complete modelling pipeline rather than as independent fixed choices. The feature-stability analysis further showed that strong predictive performance did not imply a completely stable individual feature set across patient subsets.

Overall, the findings identify selective source integration with patient-aware model development

as a promising strategy for this high-dimensional classification setting. However, the selected ADC–Post1–Pre pipeline remains a candidate model derived from a limited cohort. Independent validation in larger patient cohorts is required to determine whether its predictive performance, feature-selection pattern, and model-interpretation results are reproducible and generalisable.